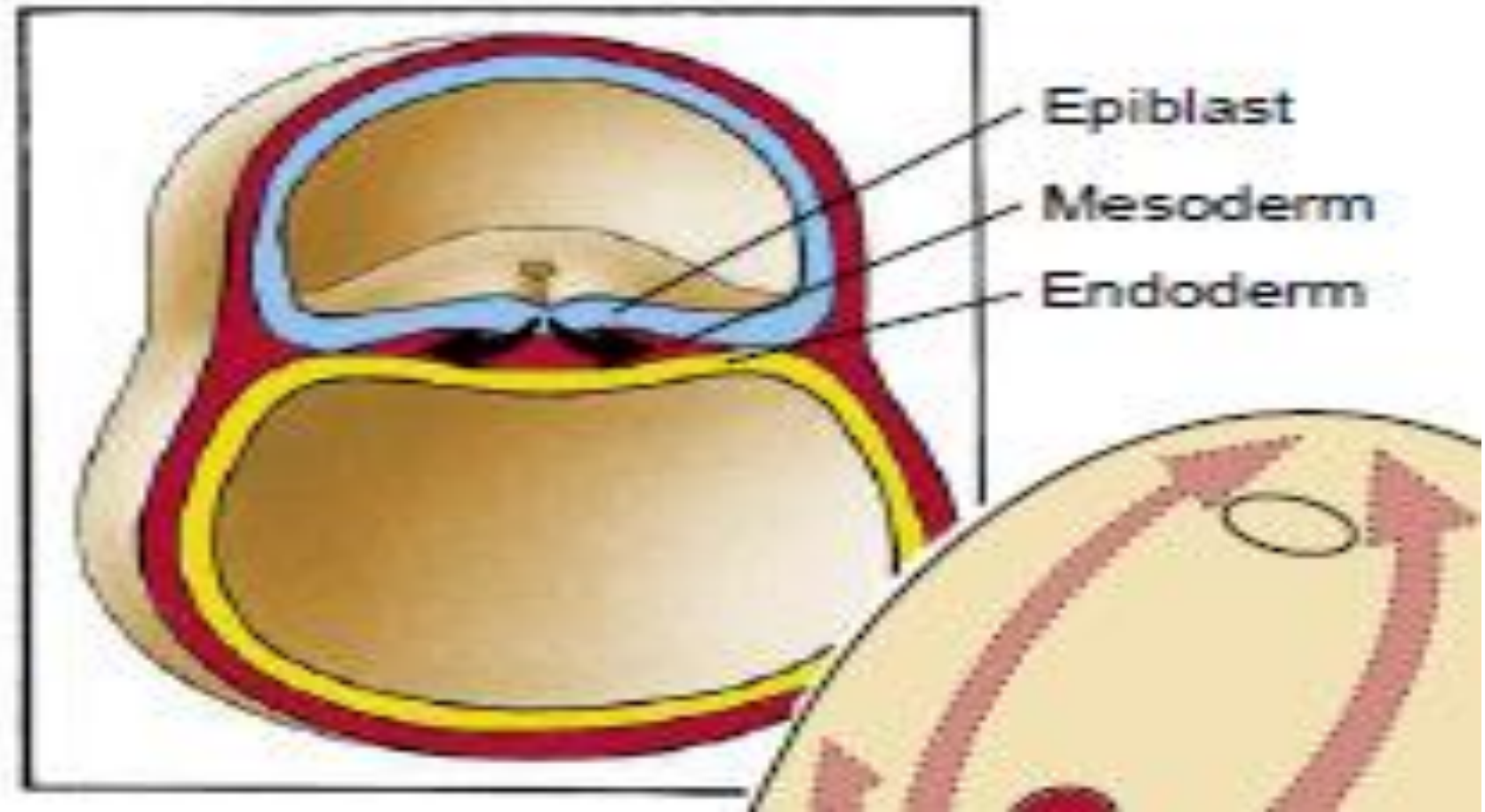


4th to 8th week of development

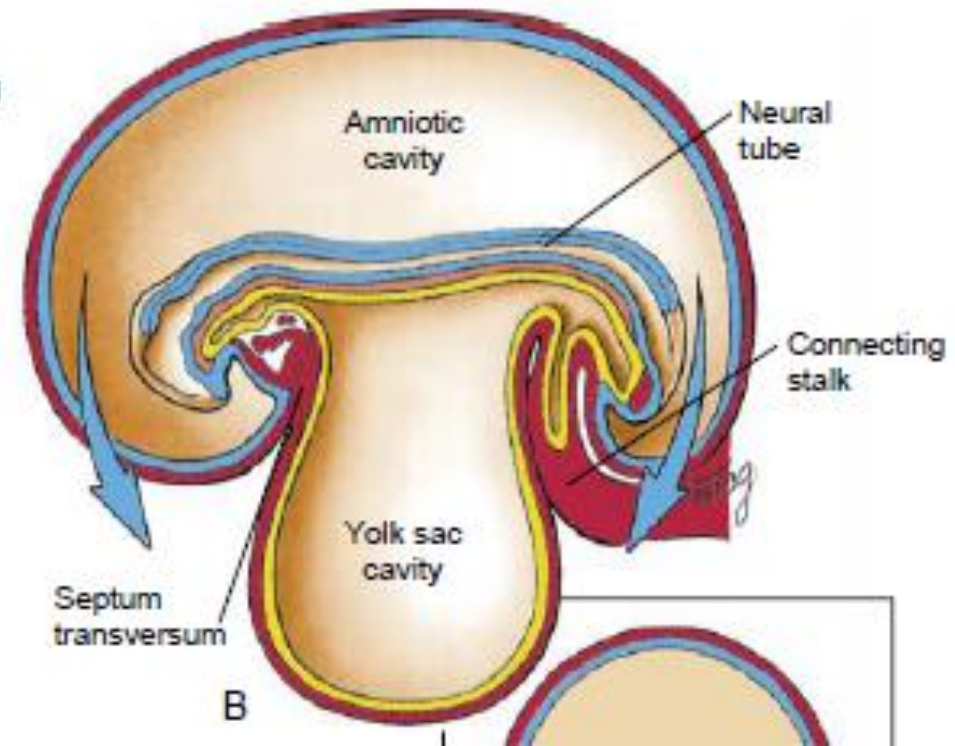
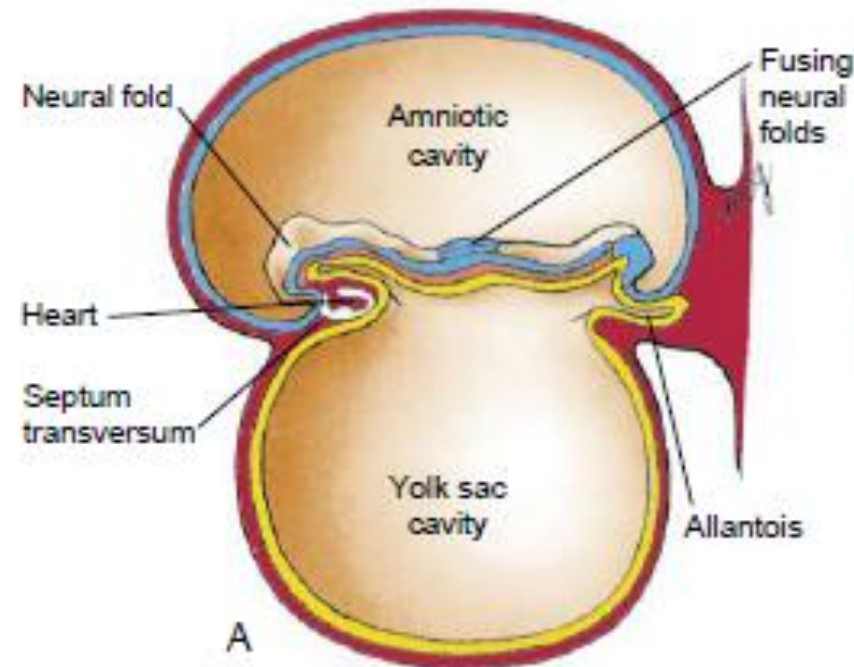
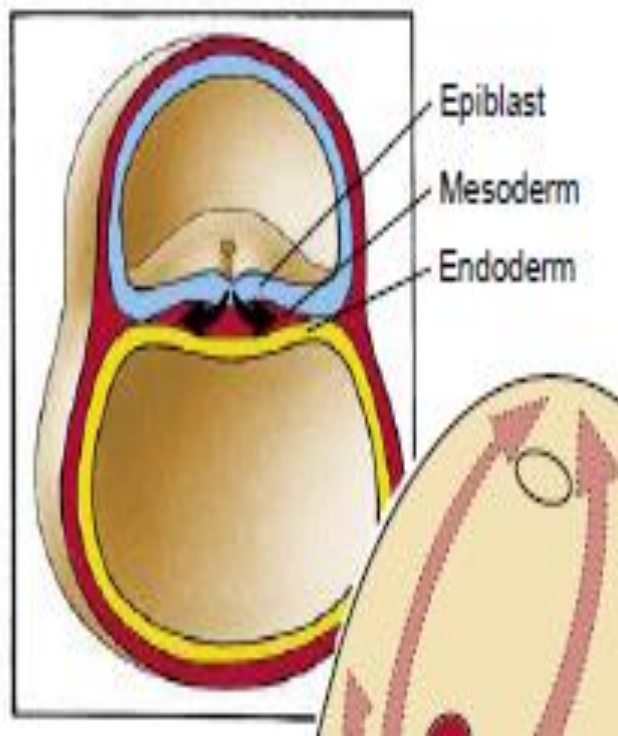
D. Ndhlovu- Chikwanda

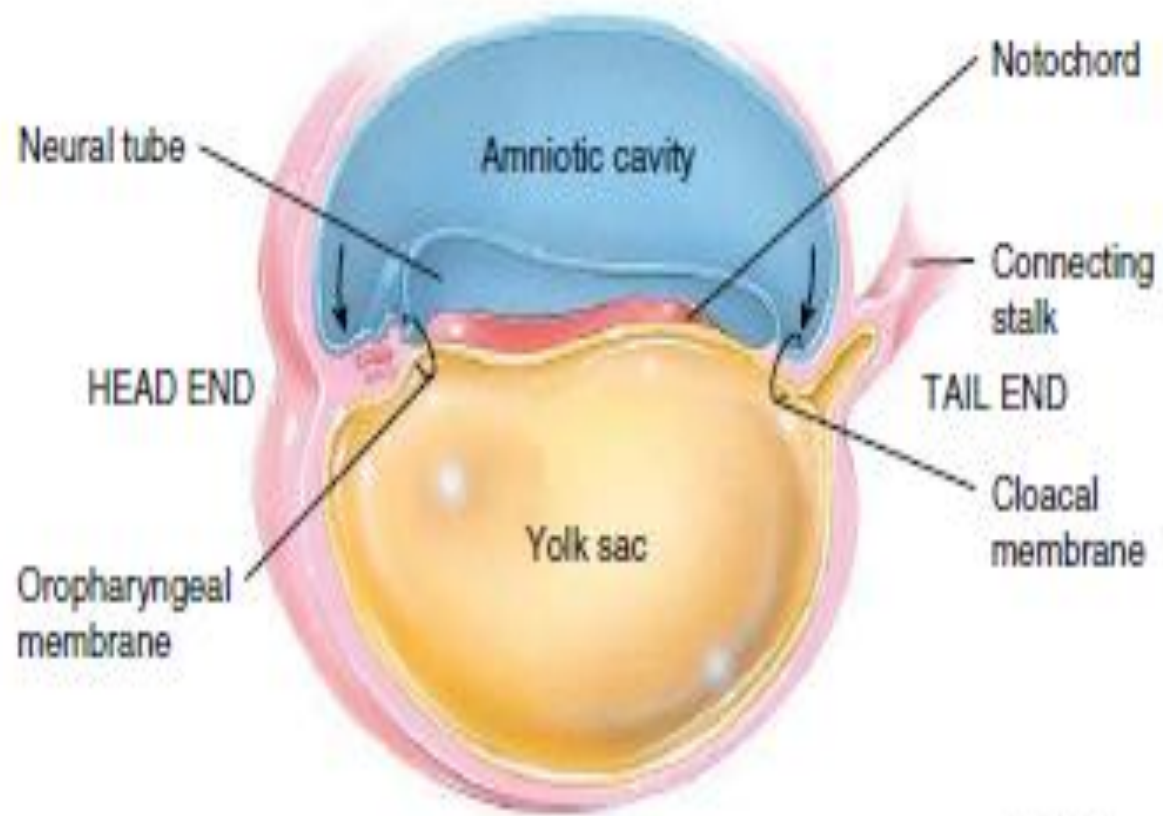
- All major external & internal structures are established during the 4th to 8th weeks
- By the end of this period, the main organ systems have begun to develop
- function of most of them is minimal except for the cardiovascular system
- As the tissues and organs form, the shape of the embryo changes, and by the 8th week, it has a distinctly human appearance
- Tissues & organs differentiate rapidly, exposure of embryos to teratogens during this period may cause major congenital anomalies
- **Teratogens** are agents such as drugs & viruses that produce or increase the incidence of congenital anomalies

- organogenesis refers to the formation of body organs and systems
- Organogenesis requires the presence of blood vessels to supply developing organs with oxygen and other nutrients
- endothelial cells of blood vessels provide a developmental signal, either a secreted substance or a direct cell-to-cell interaction, that is necessary for organogenesis

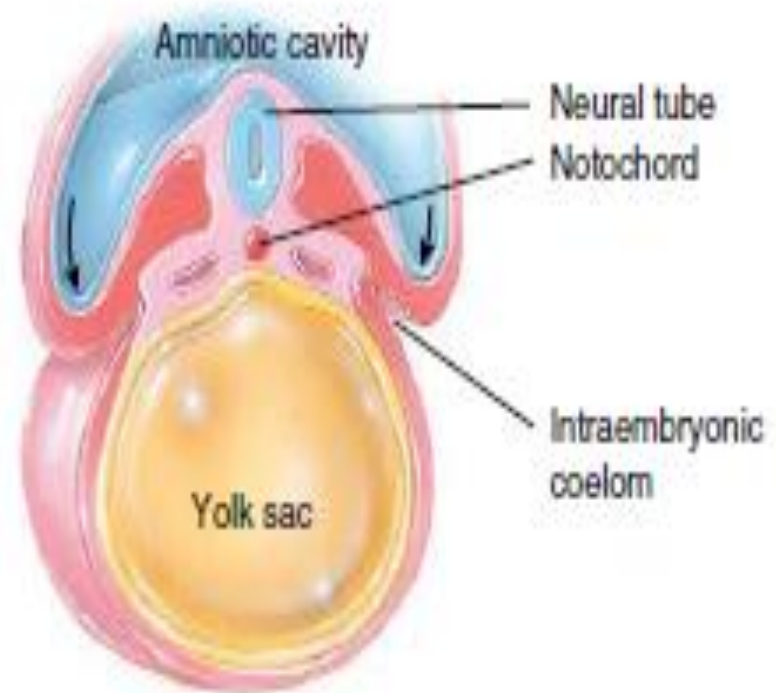


- 4th week embryo undergoes dramatic changes in shape & triples its size
- converted from a flat, two-dimensional trilaminar embryonic disc to a three-dimensional cylinder, via a process called **embryonic folding**
- cylinder consists of endoderm in the center (gut lining), ectoderm primarily on the outside (epidermis) & mesoderm in between
- cylinder does contain some internal ectoderm from neurulation (the nervous system & most of the structures of the head)

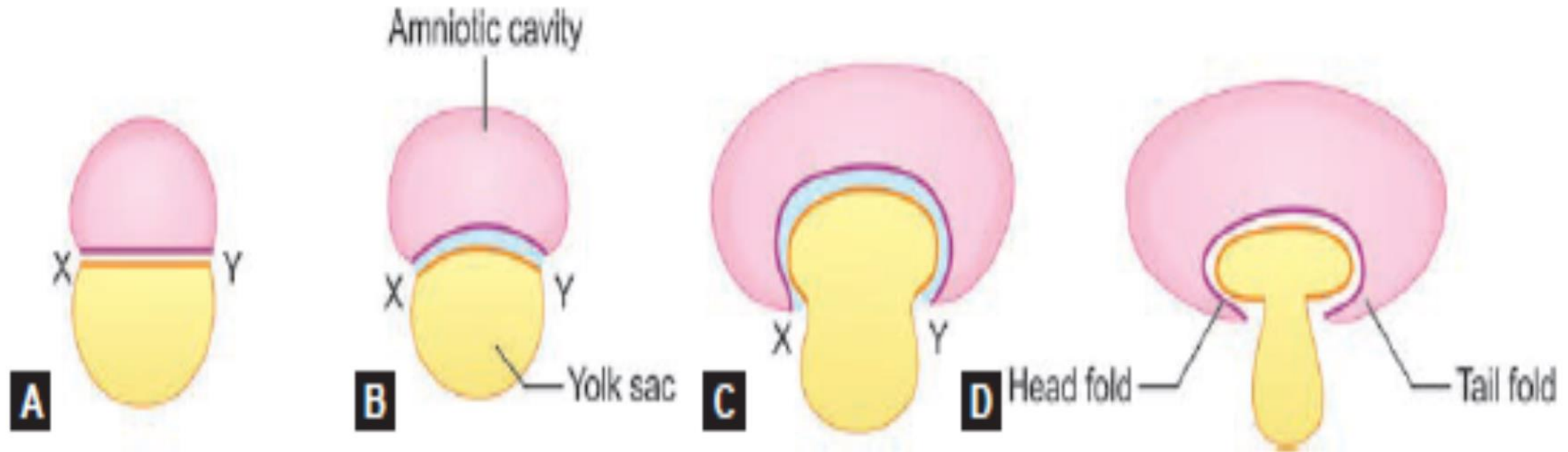




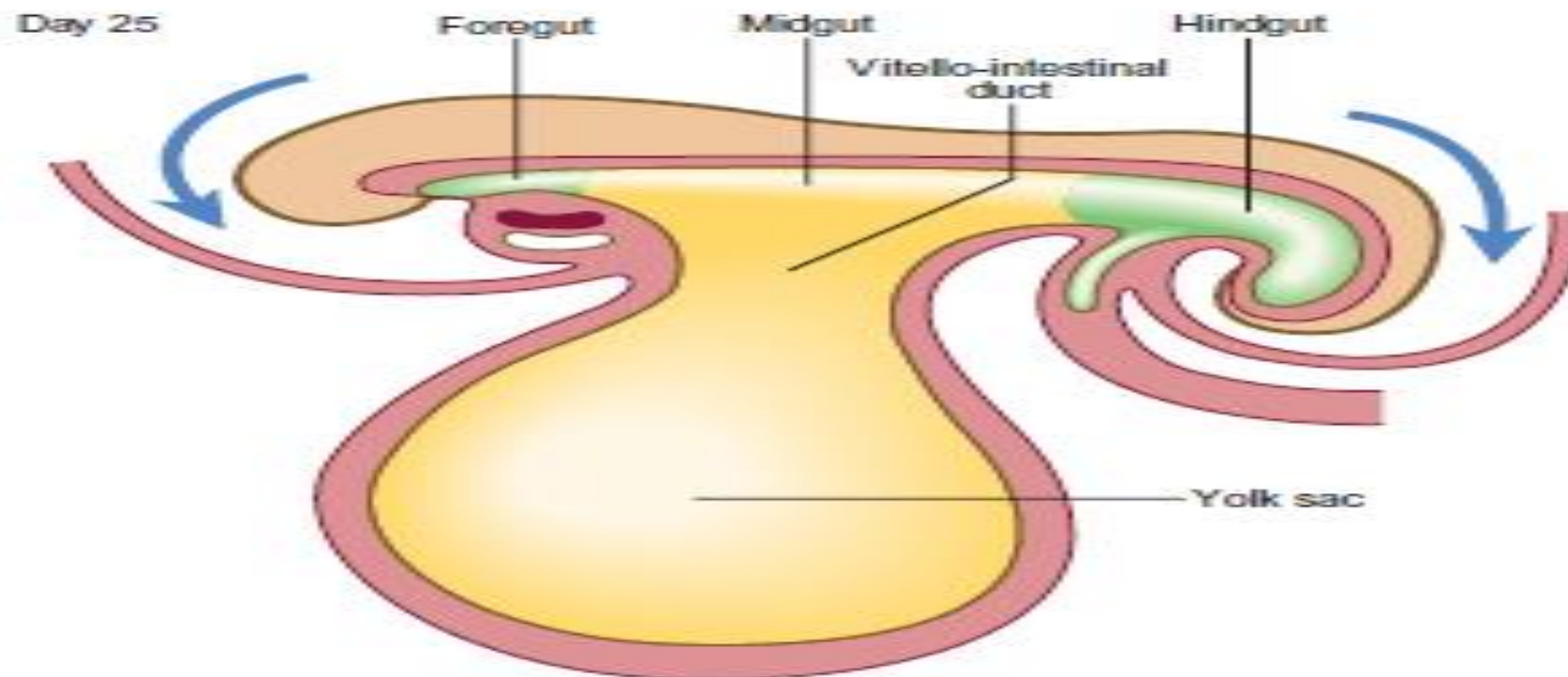
(a) 22 days



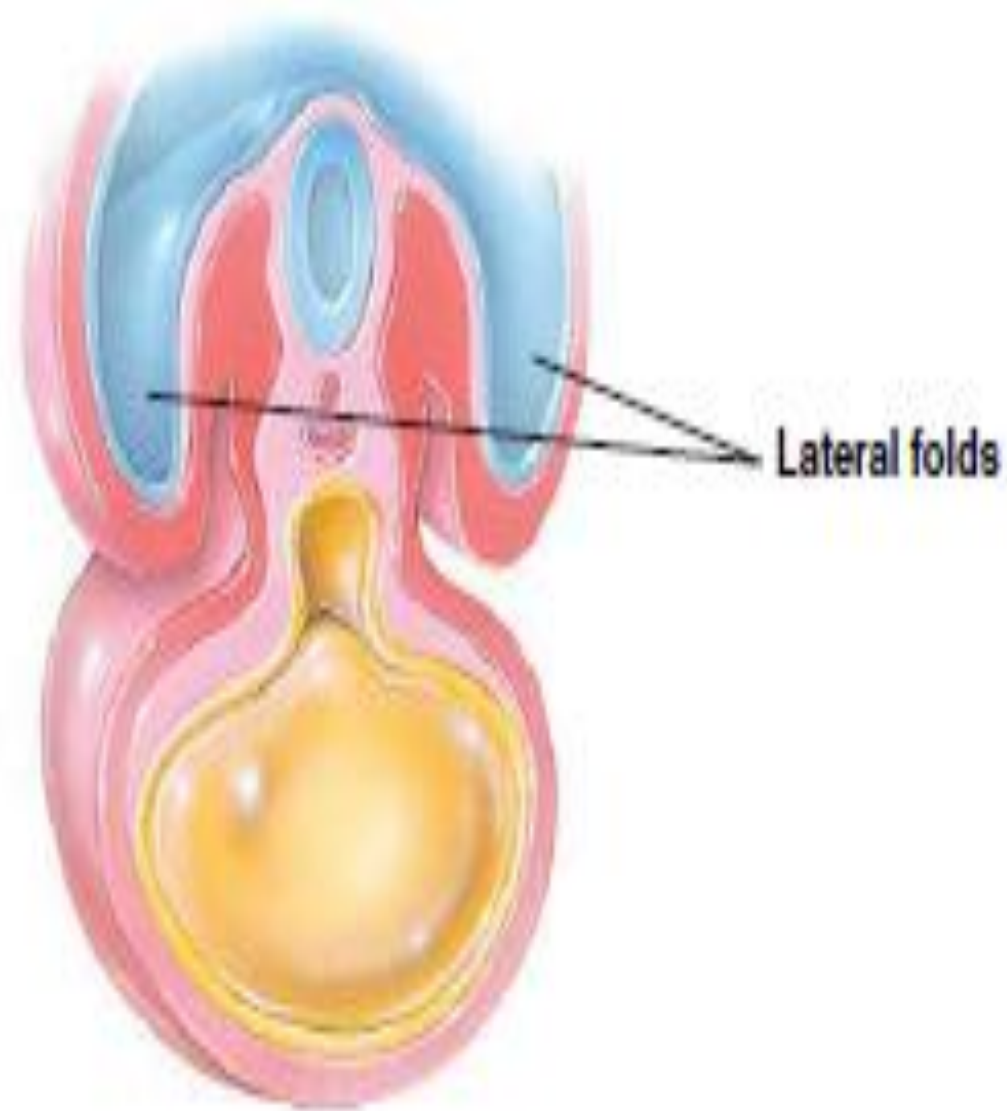
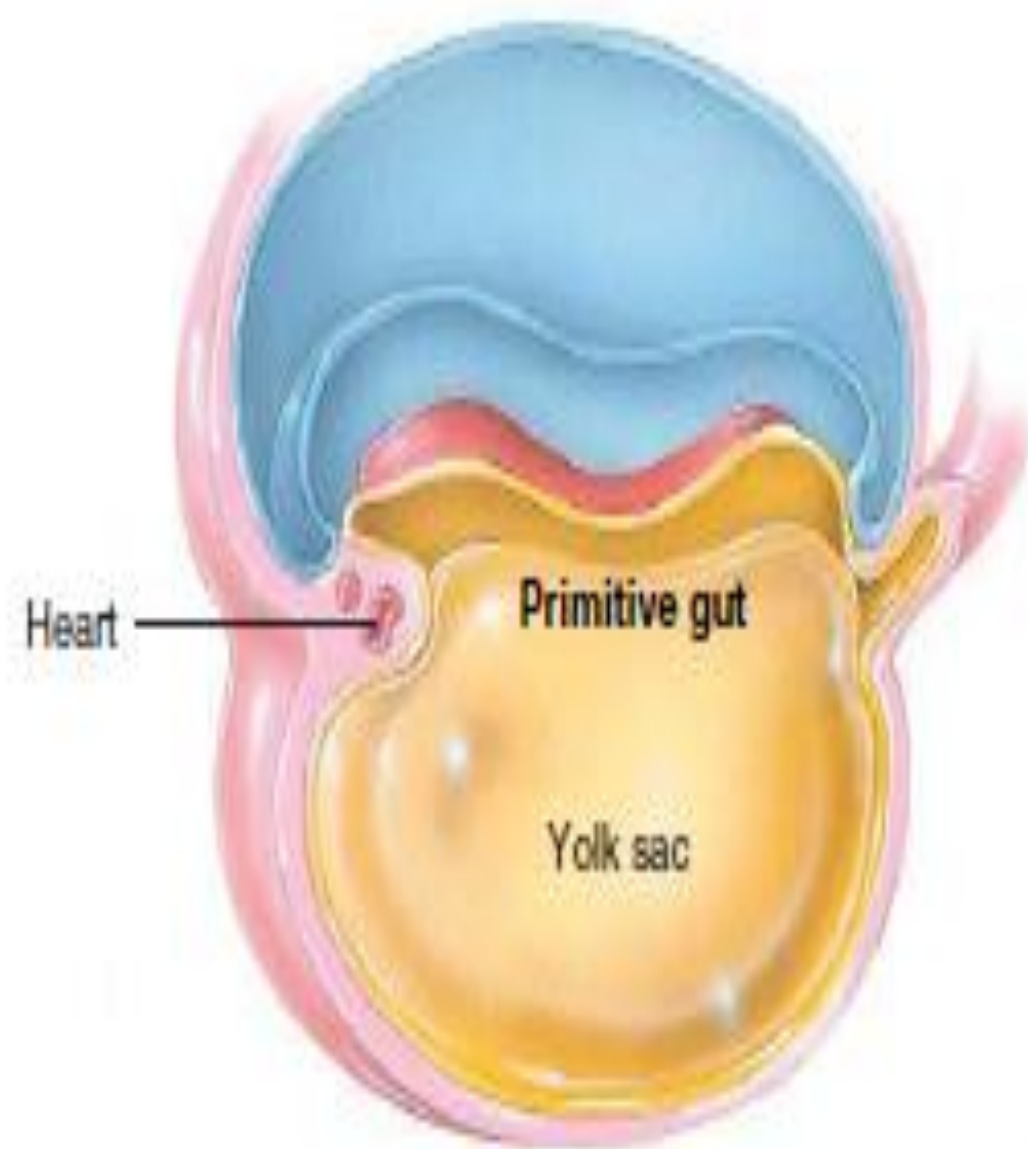
- With further enlargement, edges of embryonic disc become folded on itself in the median & transverse planes
 - folding in the median plane form ventrally directed **head fold & tail fold**
 - folding in the transverse plane forms ventrally directed **lateral folds**
 - Cephalocaudal folding is due to rapid & longitudinal growth of CNS
 - Lateral foldings are due to rapid growth of somites & convert embryo into a tubular structure
-



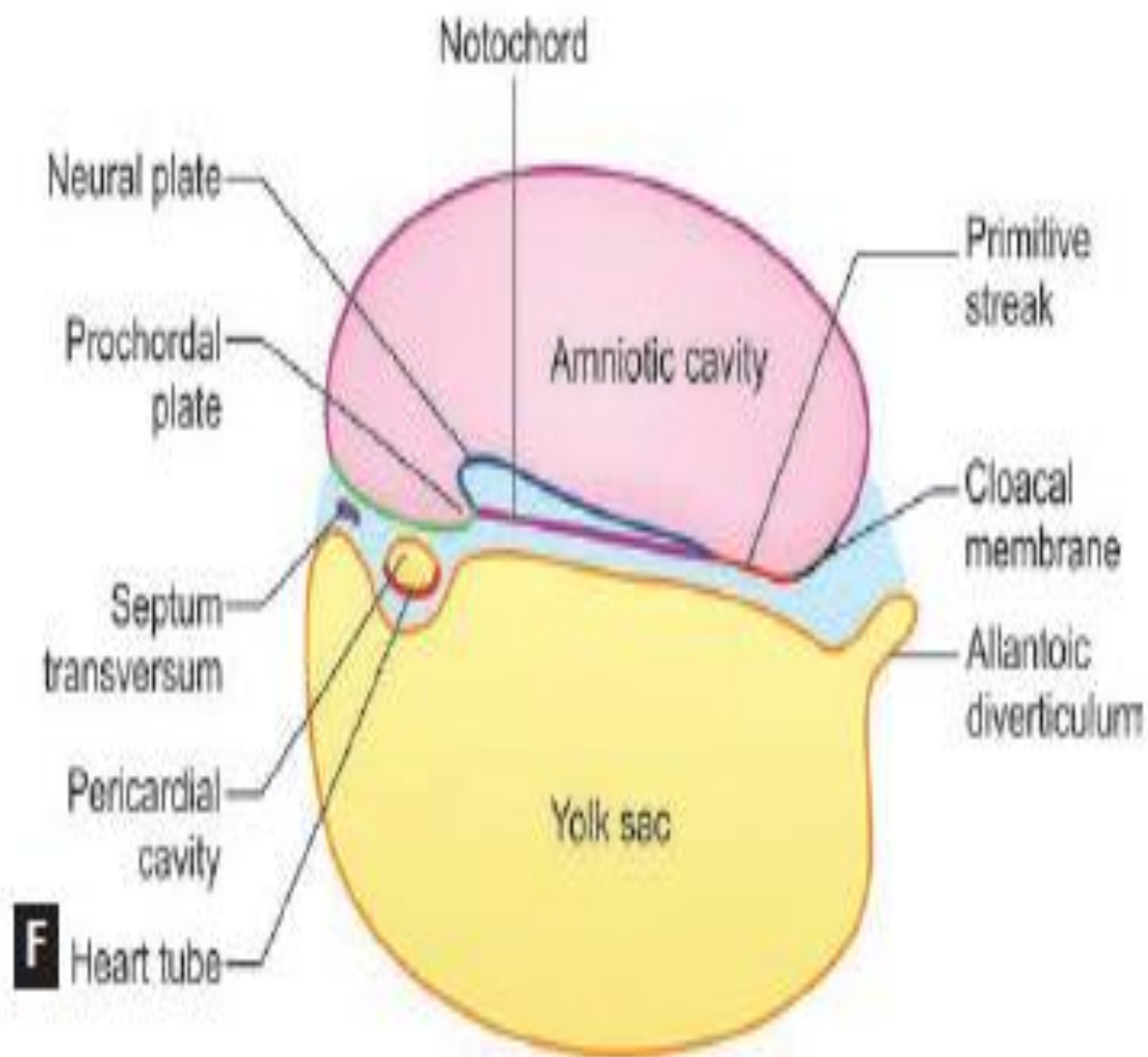
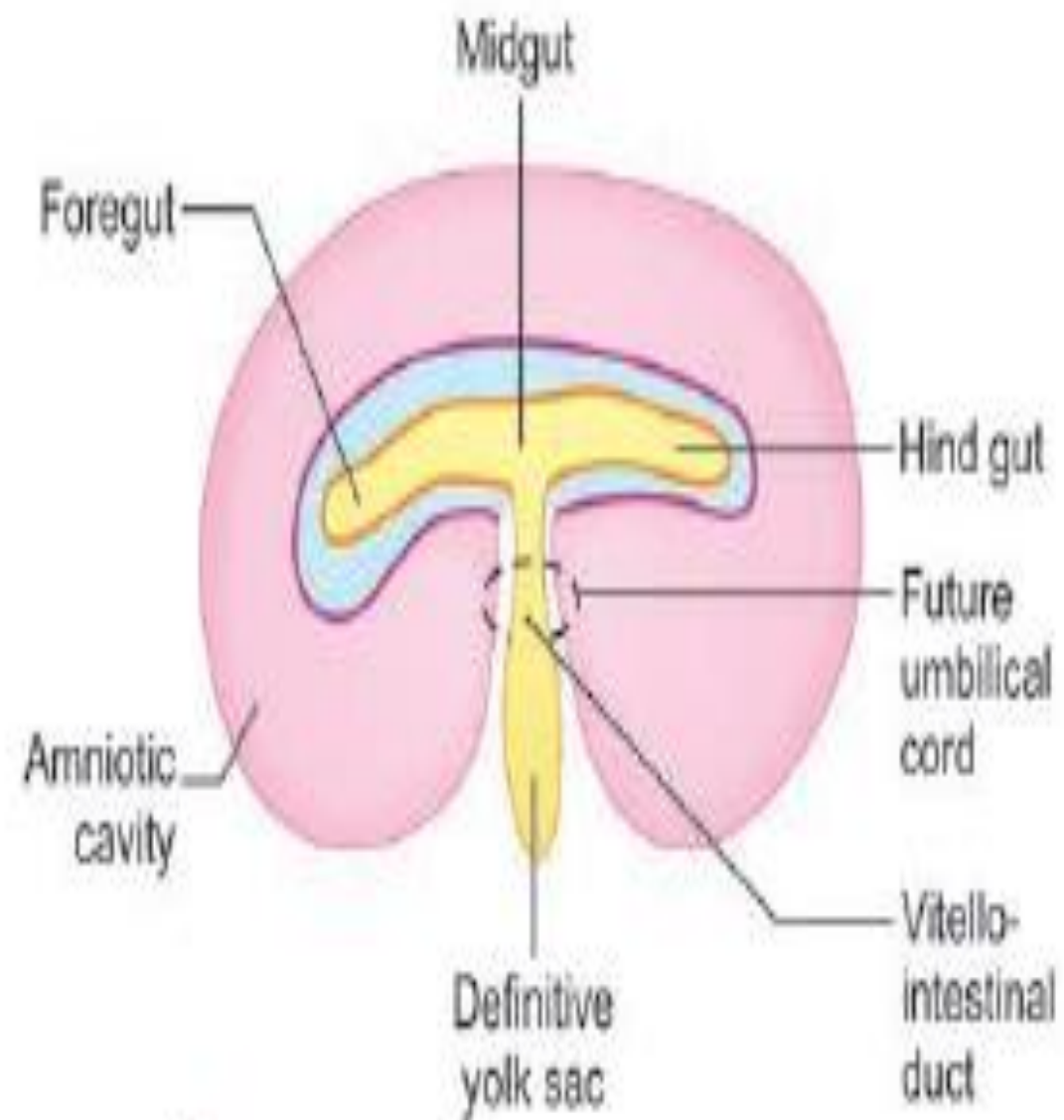
- folds occur simultaneously & merge into one another
- notochord, neural tube & somites stiffen the dorsal axis of the embryo making it foldable
- With the formation of the head & tail folds, parts of the yolk sac become enclosed within the embryo
- a tube lined by endoderm is formed in the embryo
- This is the ***primitive gut***, from which most of the gastrointestinal tract is derived



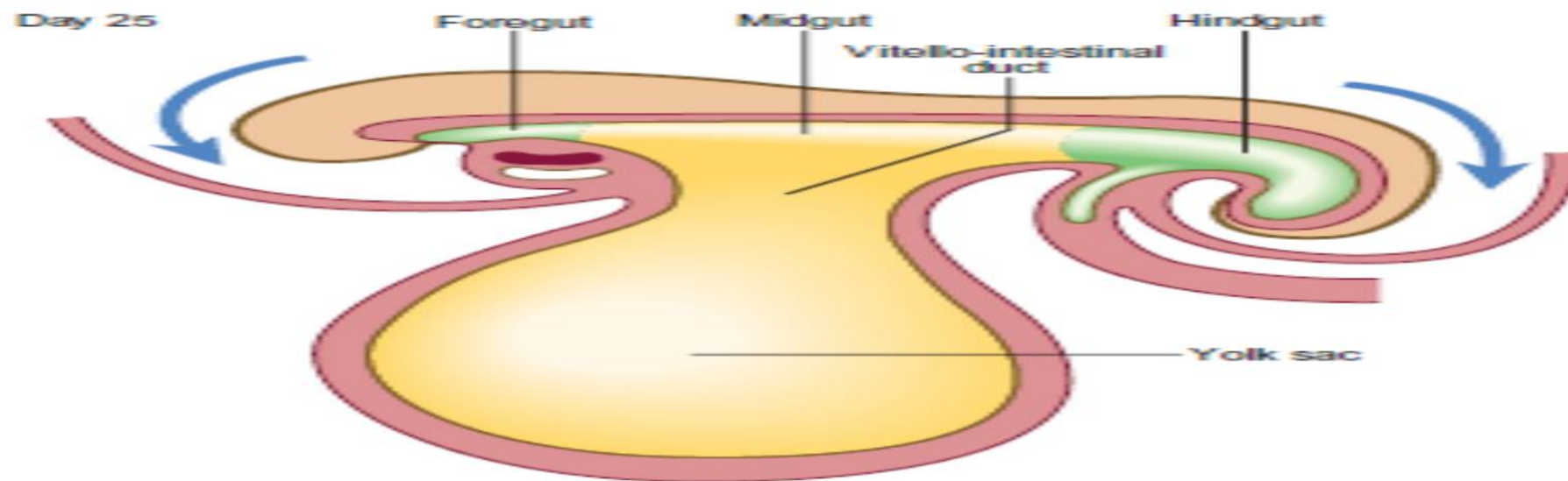
(a) 22 days



(b) 24 days



- **Midgut** is in wide communication with the yolk sac
- **foregut** is part of the gut cranial to this communication
- **hindgut** is part caudal to the communication
- communication with the yolk sac becomes progressively narrower
- Due to these changes, the yolk sac becomes small & inconspicuous, is now termed **definitive yolk sac** (also called the *umbilical vesicle*)
- narrow channel connecting it to the gut is called the **vitellointestinal duct** (also called *vitelline duct*; *yolk stalk* or *omphalomesenteric duct*)



• **foregut** derivatives are :-

A. primordial pharynx & its derivatives

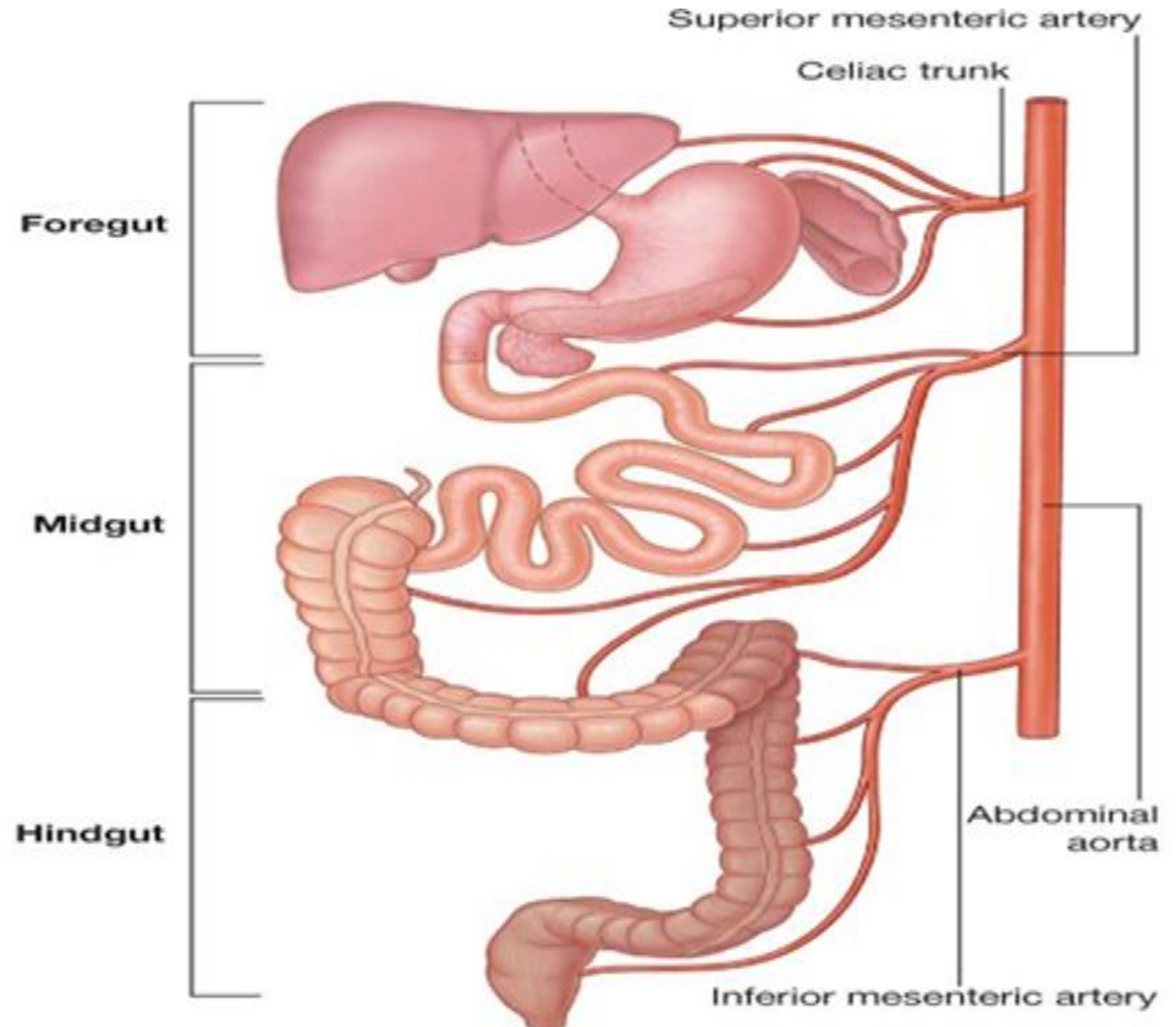
B. lower respiratory system

C. esophagus & stomach

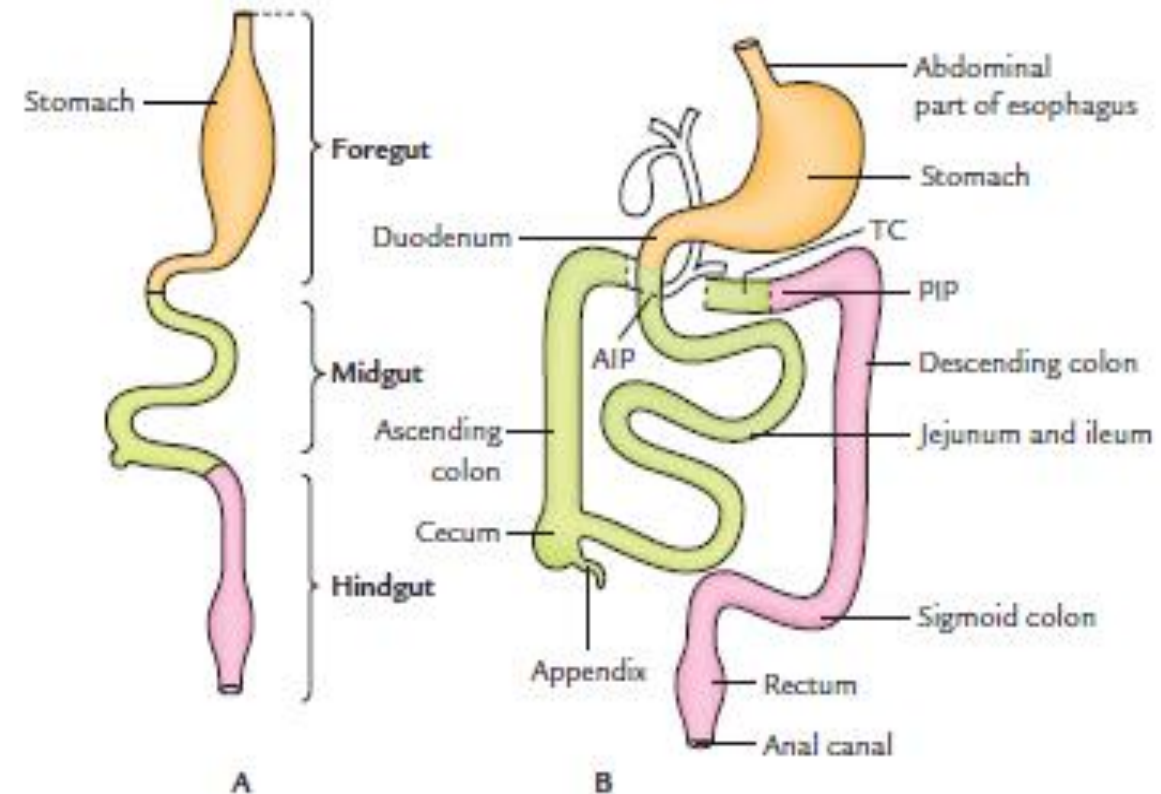
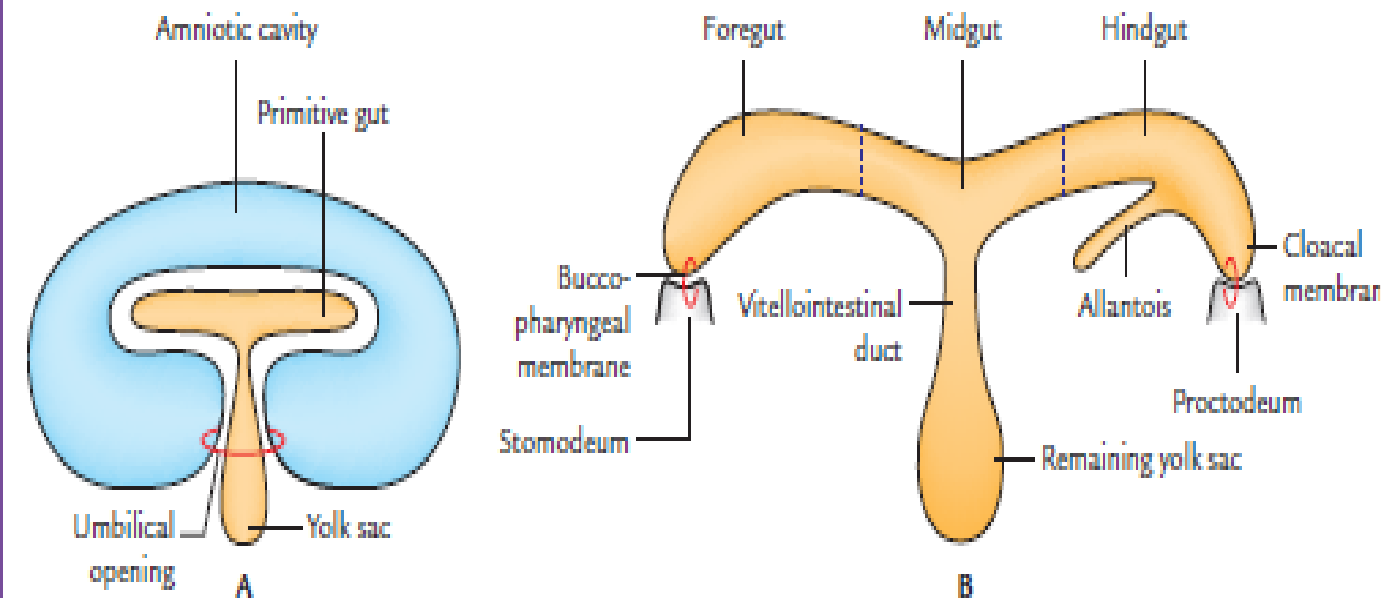
D. duodenum, proximal
to the opening of the bile duct

E. liver, gallbladder & pancreas

Supplied by the **celiac Artery**

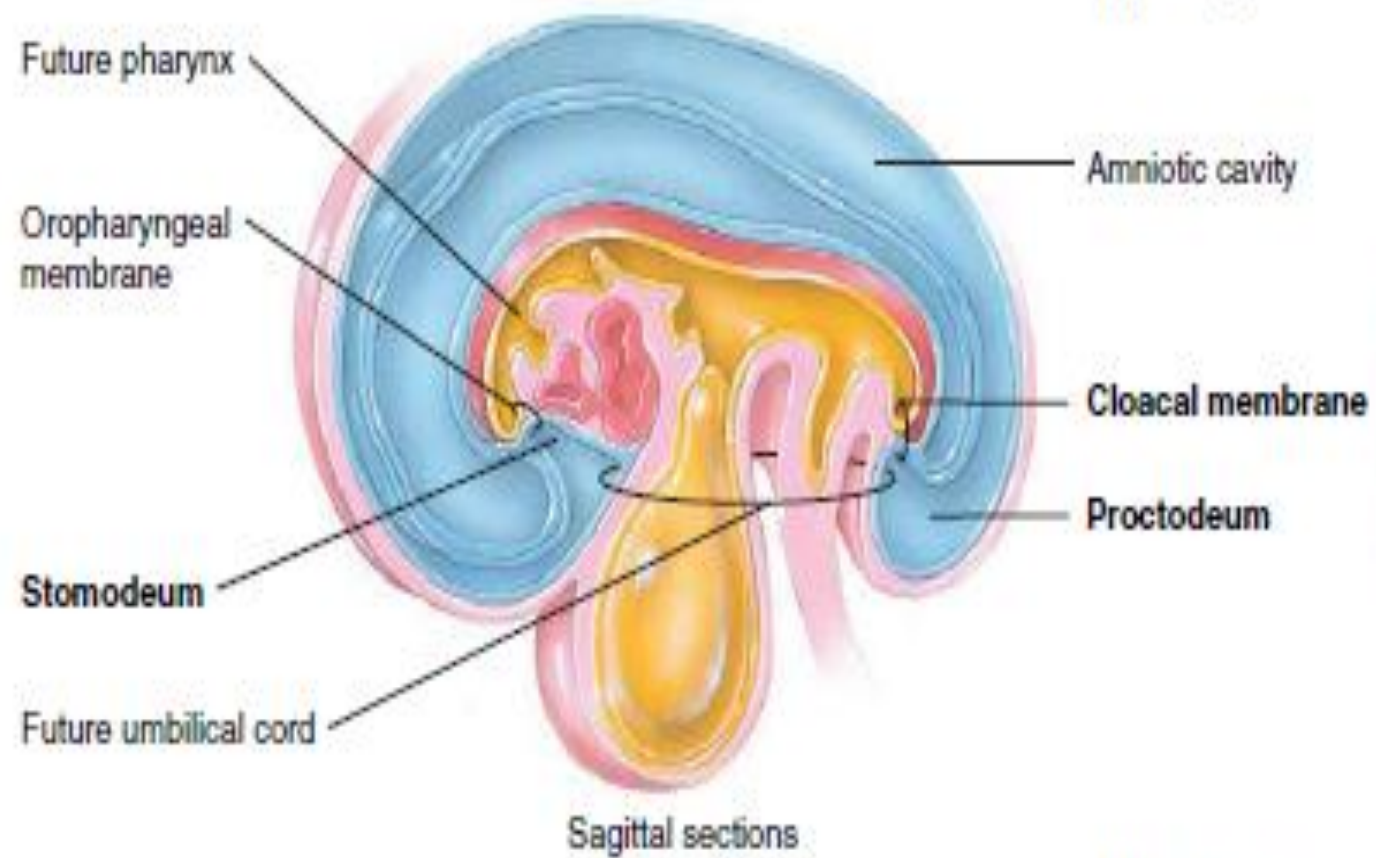


- **midgut** derivatives :-
- duodenum distal to the opening of the bile duct, jejunum, ileum
- cecum, appendix, ascending colon & right two thirds of the transverse colon
- supplied by **superior mesenteric artery**
- **hindgut** derivatives are:-
- left one third to one half of t transverse colon, descending colon & sigmoid colon, rectum, and superior part of the anal canal
- Supplied by **Inferior mesenteric artery**

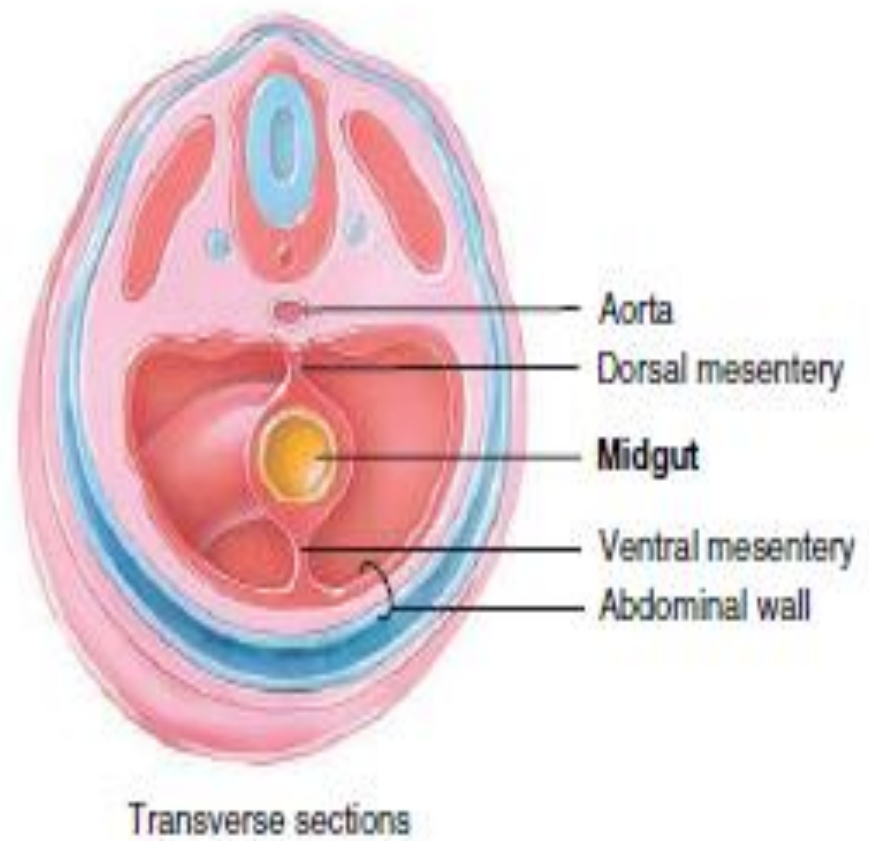


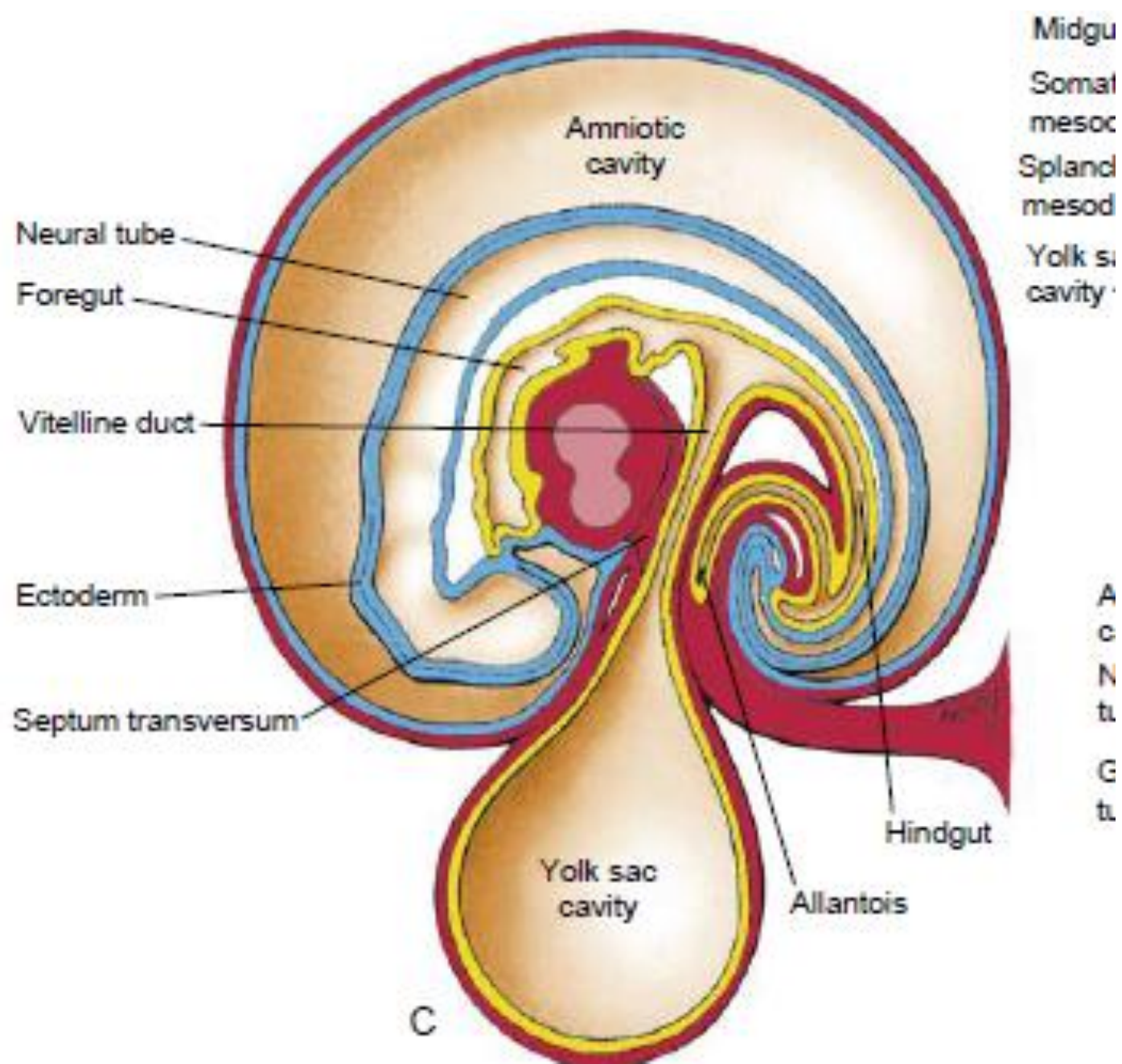
- As the head & tail folds forms, folds are also formed on each side in transverse or horizontal plane are the ***lateral folds***
- Results in embryo being enclosed all around by ectoderm except in the region through which the *vitellointestinal duct (omphalomesenteric duct)* passes
- Forms circular aperture the ***umbilical opening***
- folding facilitates growth & expansion of amniotic cavity that comes to surround the embryo on all sides
- the embryo floats in the amniotic fluid, which fills the cavity
- Convergence of folds on ventral surface forms **tubular investment of amnion for connecting stalk**
- This causes obliteration of extraembryonic coelom and now amnion forms a covering for the umbilical cord

(c) 26 days

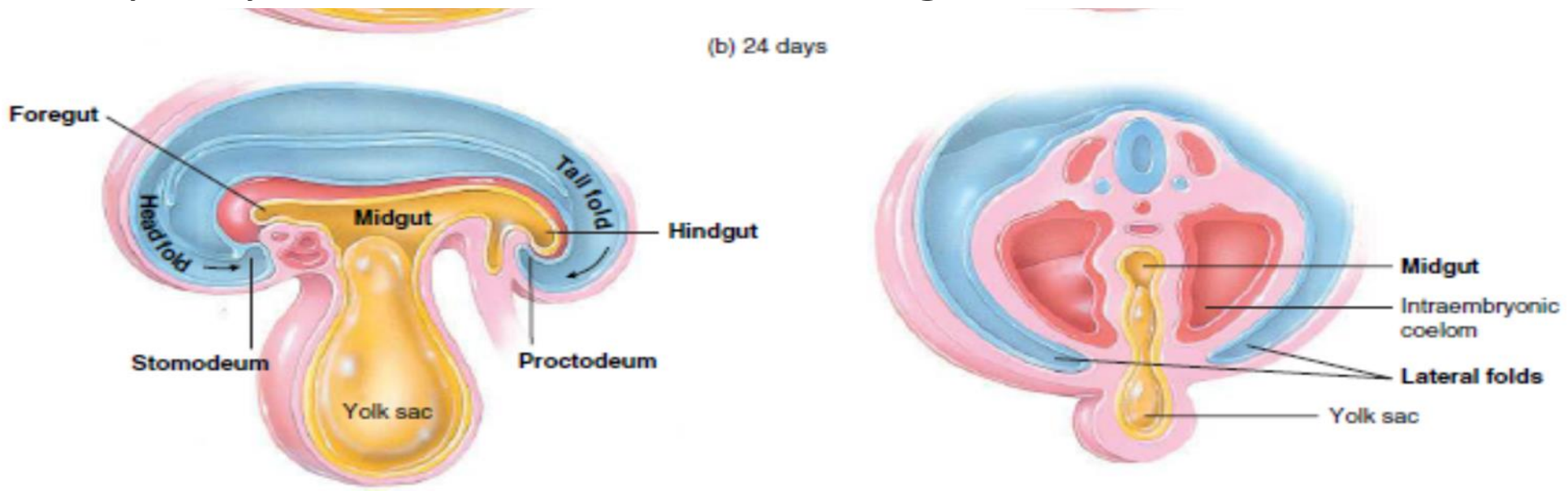


(d) 28 days

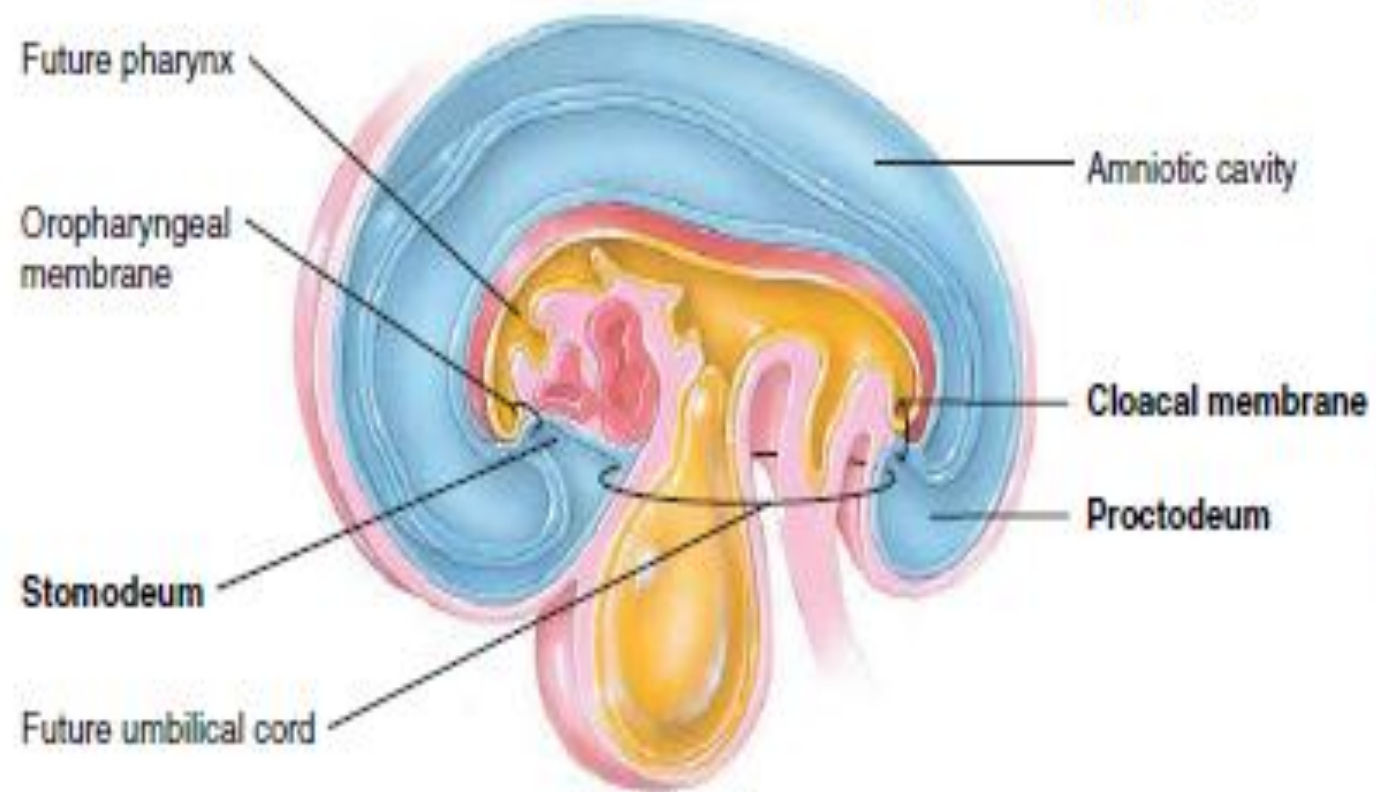




- **oropharyngeal membrane** is located in the head end of the embryo
- It separates the future pharyngeal (throat) region of the foregut from the **stomodeum** the future oral (mouth) cavity
- **stomodeum** is an ectodermal depression at the head end between the bulging head & pericardial bulge
- Because of head folding, oropharyngeal membrane moves downward & the foregut & stomodeum move closer to their final positions
- When the oropharyngeal membrane ruptures during the 4th week, pharyngeal region of the pharynx & stomodeum are brought into contact with each other

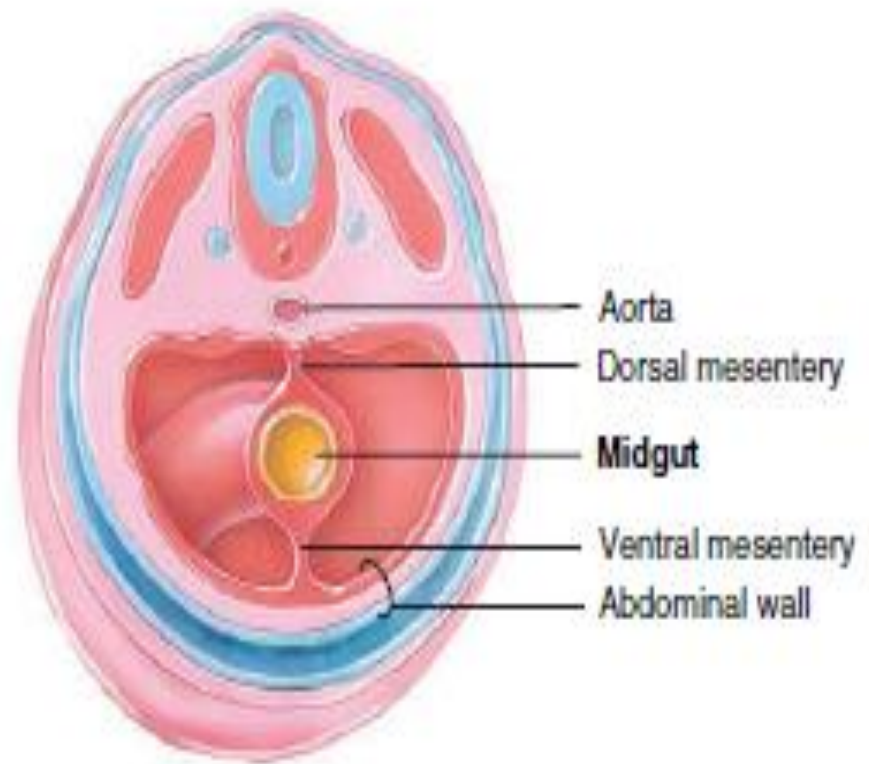


(c) 26 days



Sagittal sections

(d) 28 days



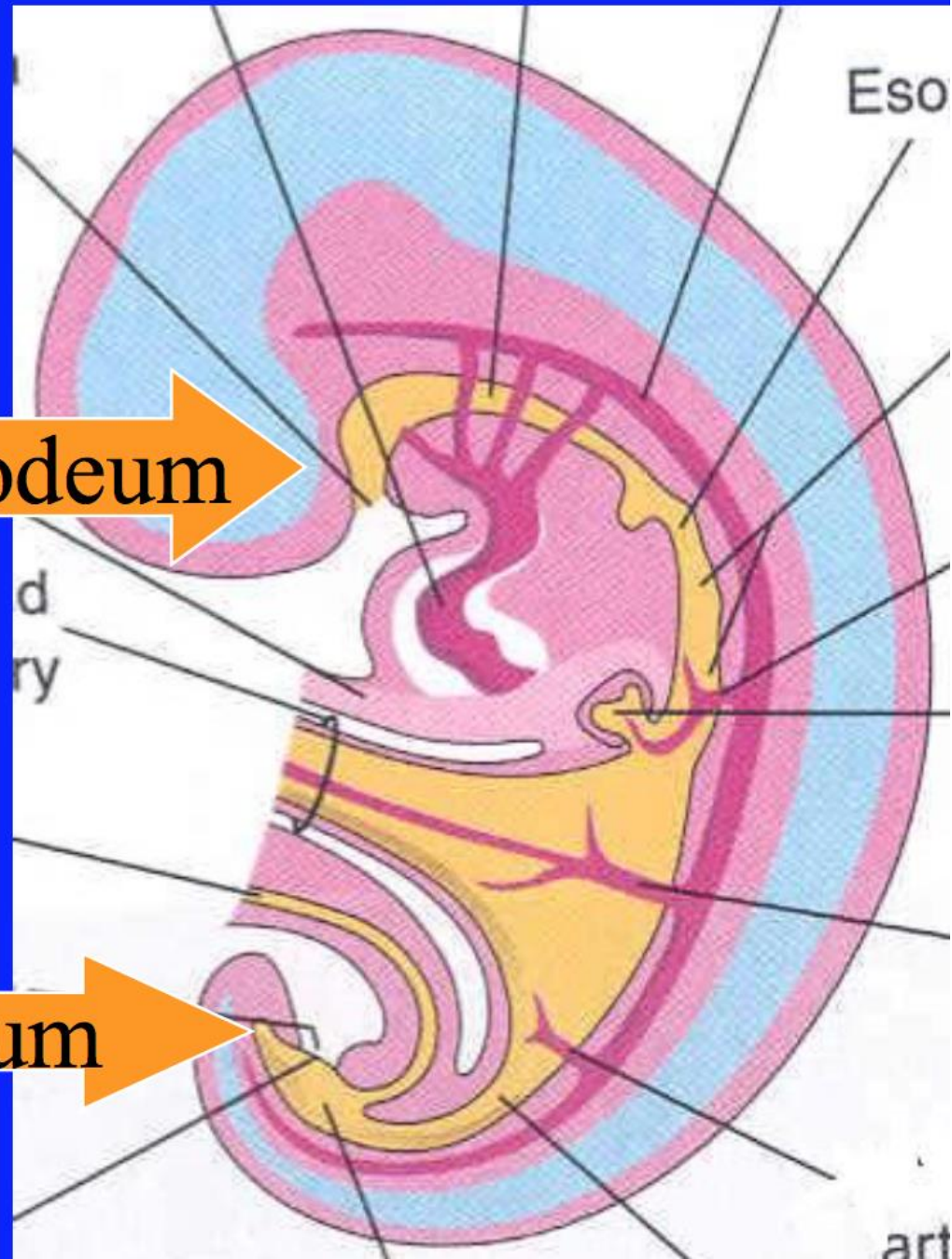
Transverse sections

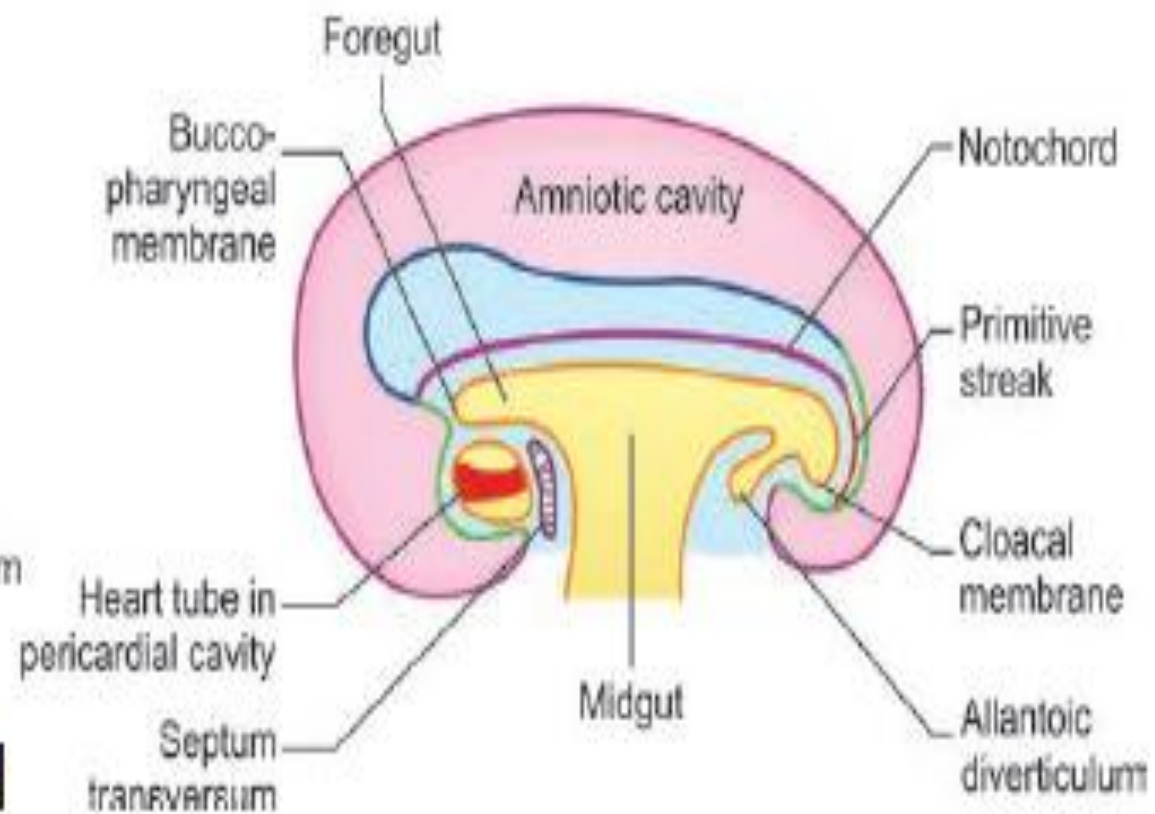
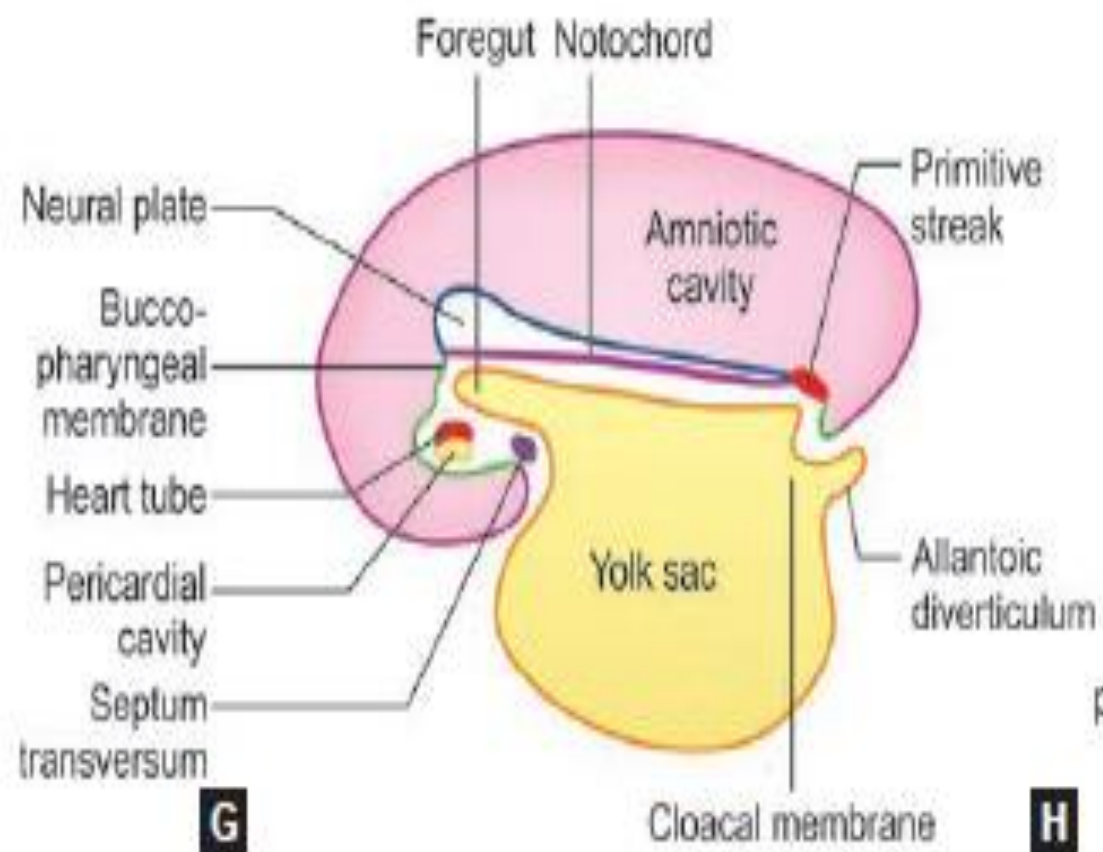
- In a developing embryo, the last part of the hindgut expands into a cavity called the **cloaca**
- On the outside of an embryo is a small cavity in the tail region called the **proctodeum**
- **cloacal membrane** separates the cloaca from the proctodeum
- During embryonic development, the cloaca divides into a **ventral urogenital sinus** and a **dorsal anorectal canal**
- As a result of tail folding, the cloacal membrane moves downward and the urogenital sinus, anorectal canal, and proctodeum move closer to their final positions.
- When the cloacal membrane ruptures during the 7th week of development, the urogenital & anal openings are created

stomodeum

no mesodermal
tissue at these two areas

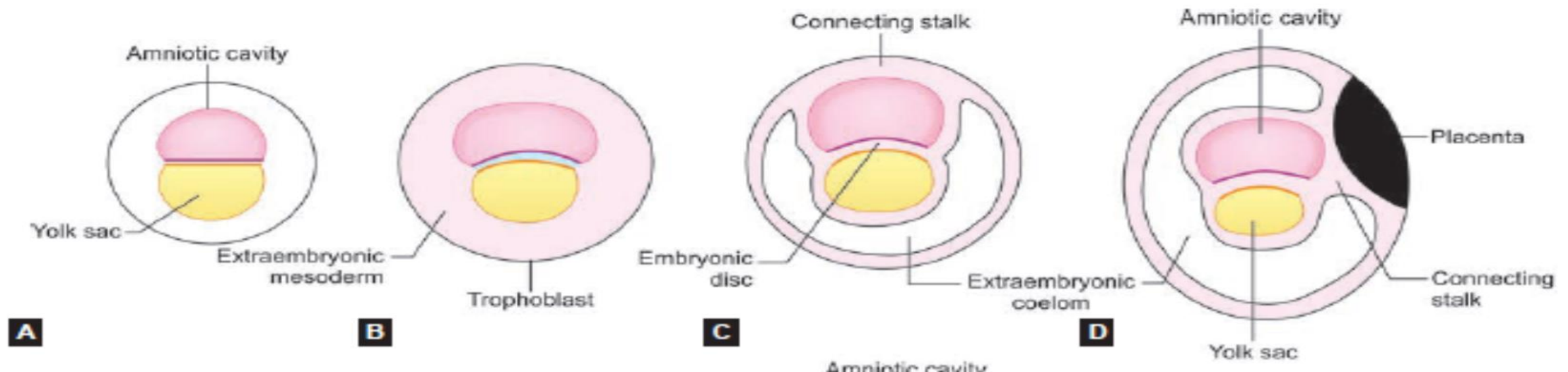
proctodeum



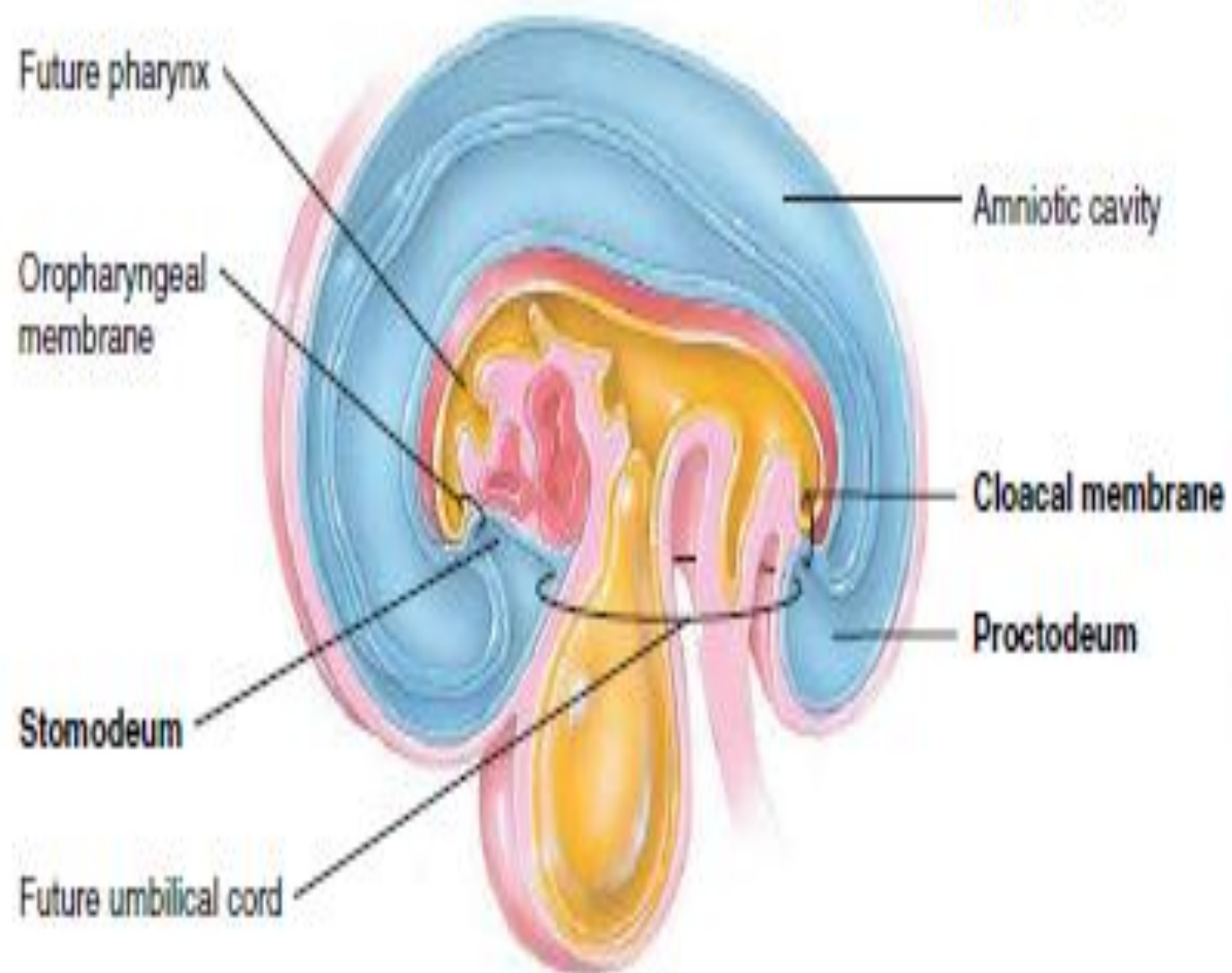


• CONNECTING STALK

- With the formation of the tail fold, the attachment of the connecting stalk moves (with the tail end of the embryonic disc) to the ventral aspect of the embryo
- It is now attached in the region of the umbilical opening
- blood vessels have developed in the embryo & also in the placenta
- These sets of blood vessels are in communication by means of arteries & veins passing through the connecting stalk
- At first, there are 2 arteries & 2 veins in the connecting stalk, but later the right vein disappears (the left vein is “left”)

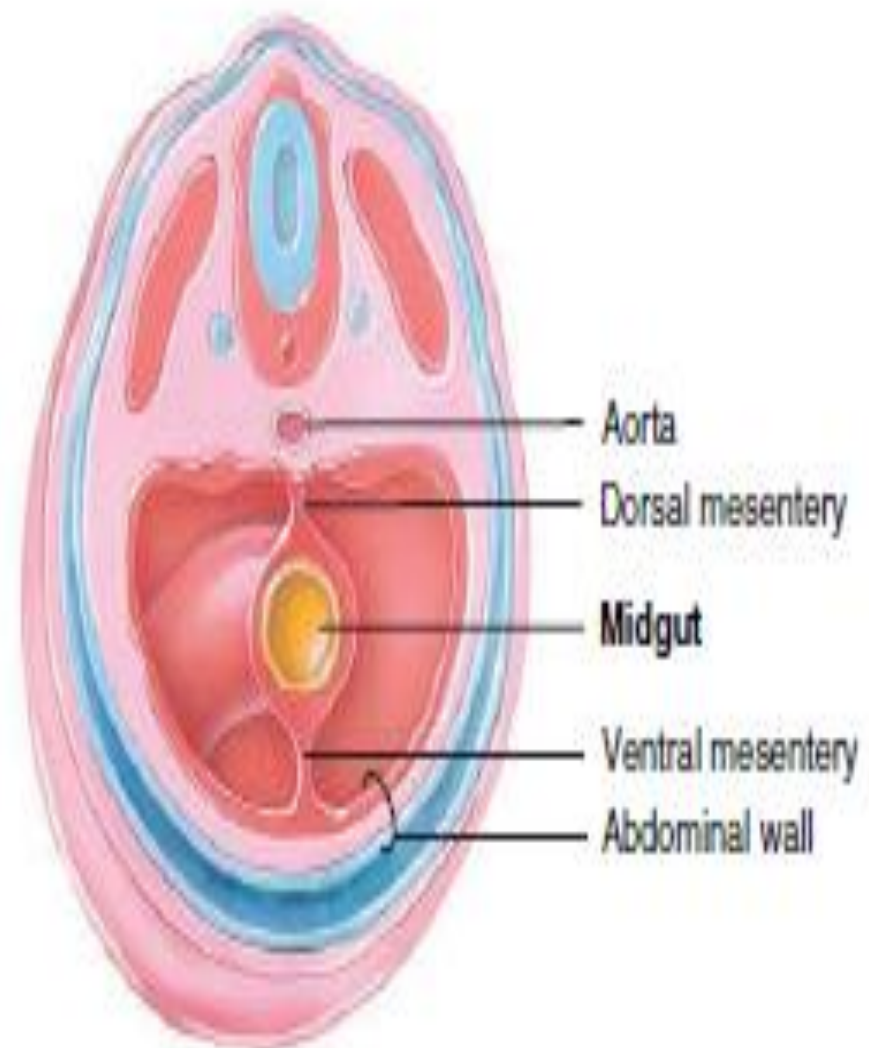


(c) 26 days



Sagittal sections

(d) 28 days



Transverse sections

- the amnion has a circular attachment to the margins of the umbilical opening & forms a wide tube in which the following lie:
 1. Vitellointestinal duct & remnants of the yolk sac
 2. Mesoderm (extraembryonic) of the connecting stalk
- mesoderm gets converted into a gelatinous substance called “**Wharton’s jelly**”
- protects blood vessels in the umbilical cord
- 3. Blood vessels that pass from the embryo to placenta
- 4. A small part of the extraembryonic coelom
- This tube of amnion & structures within it, constitutes the ***umbilical cord***

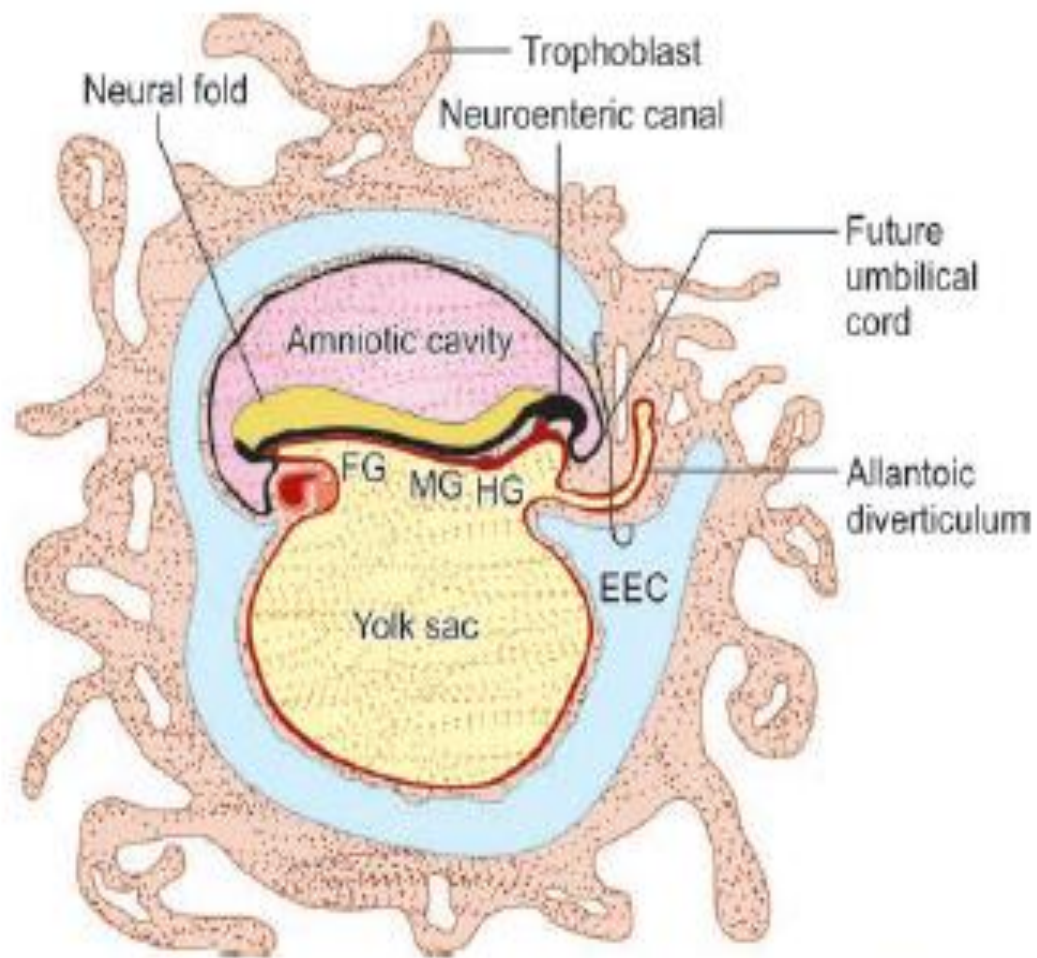


Fig. 5.12: Structures forming the umbilical cord
Abbreviations: FG: Foregut; MG: Midgut; HG: Hindgut;
 EEC: Extraembryonic coelom

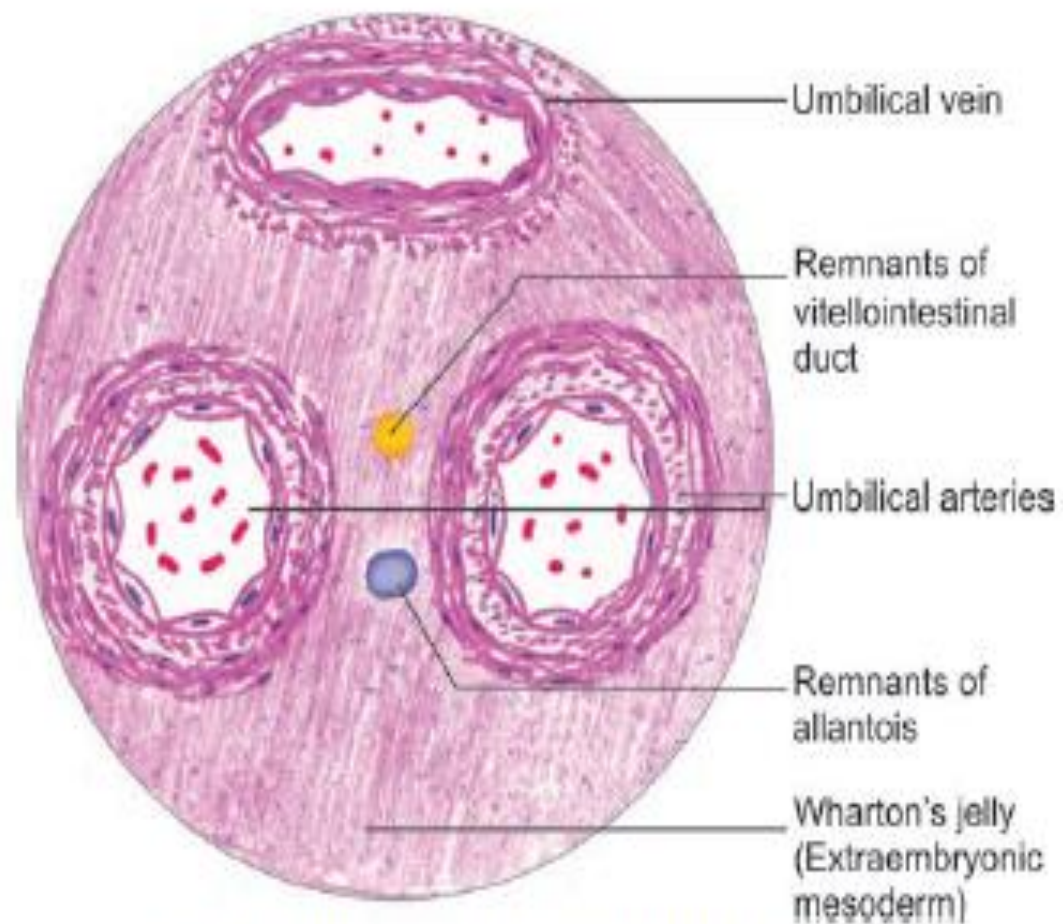


Fig. 5.13: section through umbilical cord

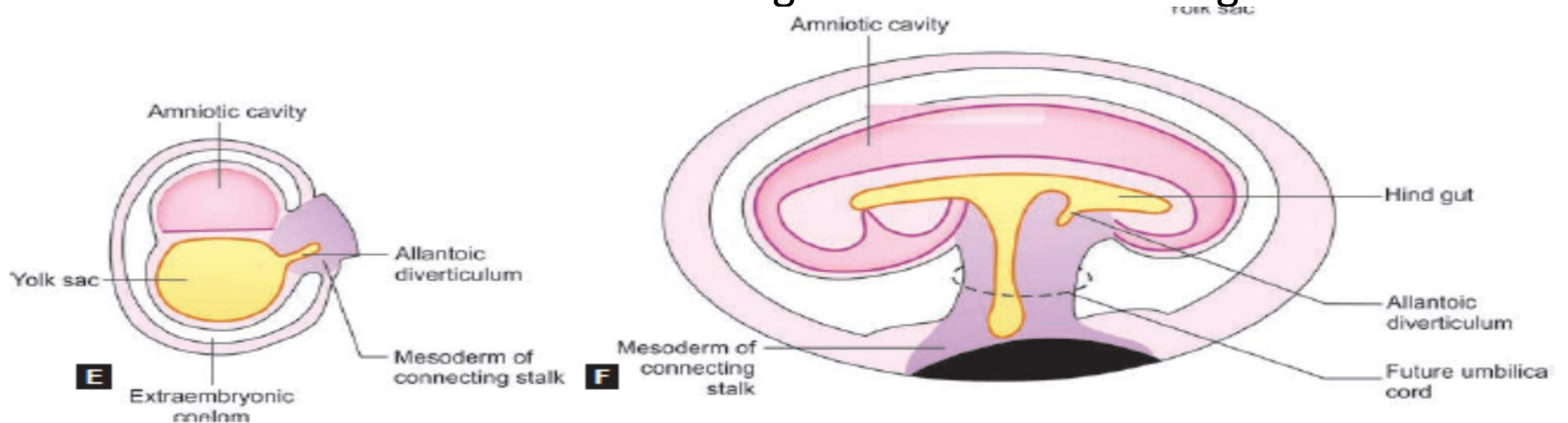
- **Clinical Correlation**

- ***Cord prolapse:***

- The umbilical cord due to its long length is likely to prolapse through the uterus during parturition (child birth) leading to a clinical condition called *cord prolapse*
- this condition, the cord is likely to be compressed between fetal head and pelvic wall of mother
- This may lead to hypoxia to the fetus
- about one-fifth of all deliveries, the cord may encircle the neck of the fetus and may cause strangulation
- Too short cord may create difficulty in parturition by pulling the placenta

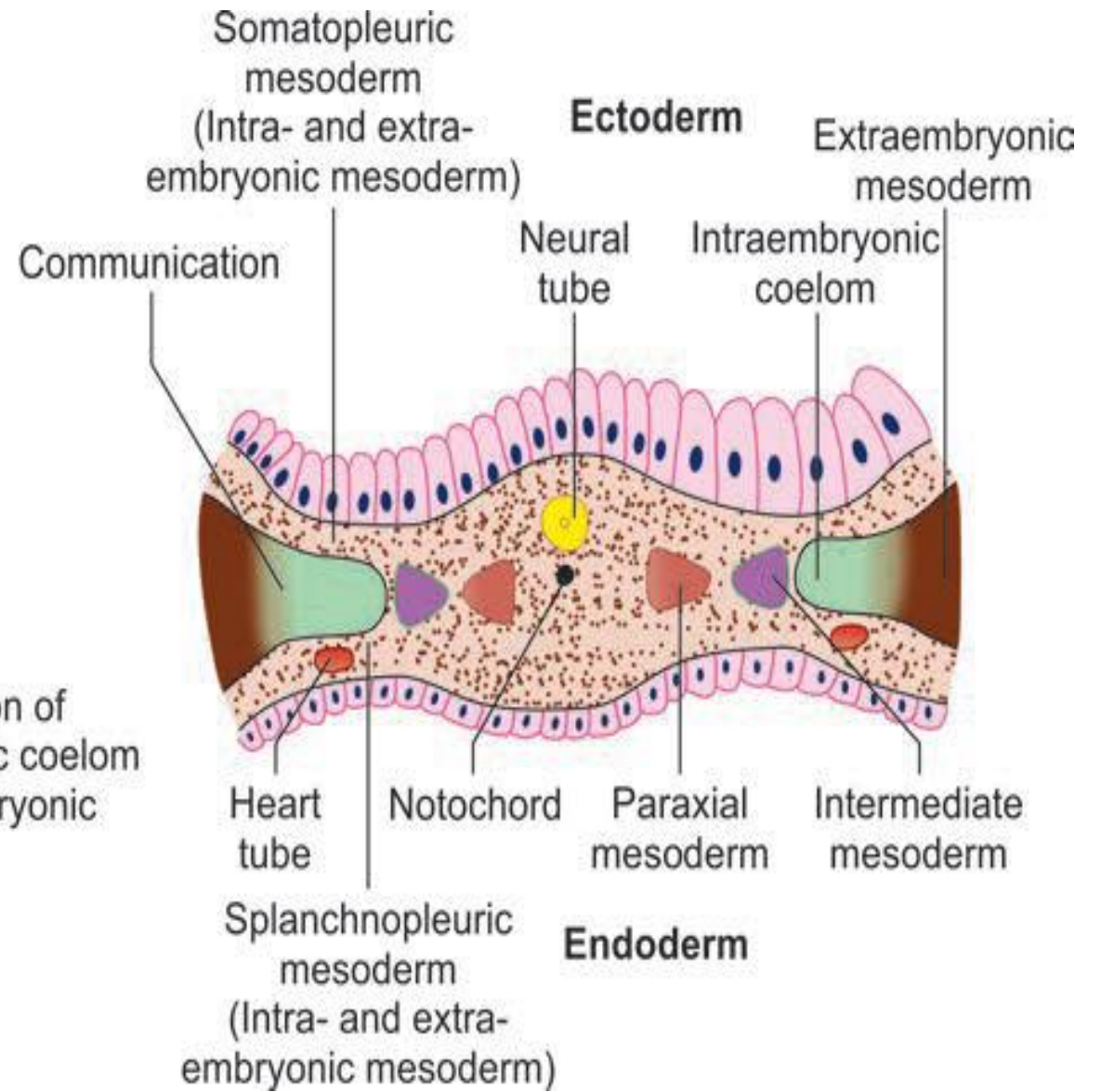
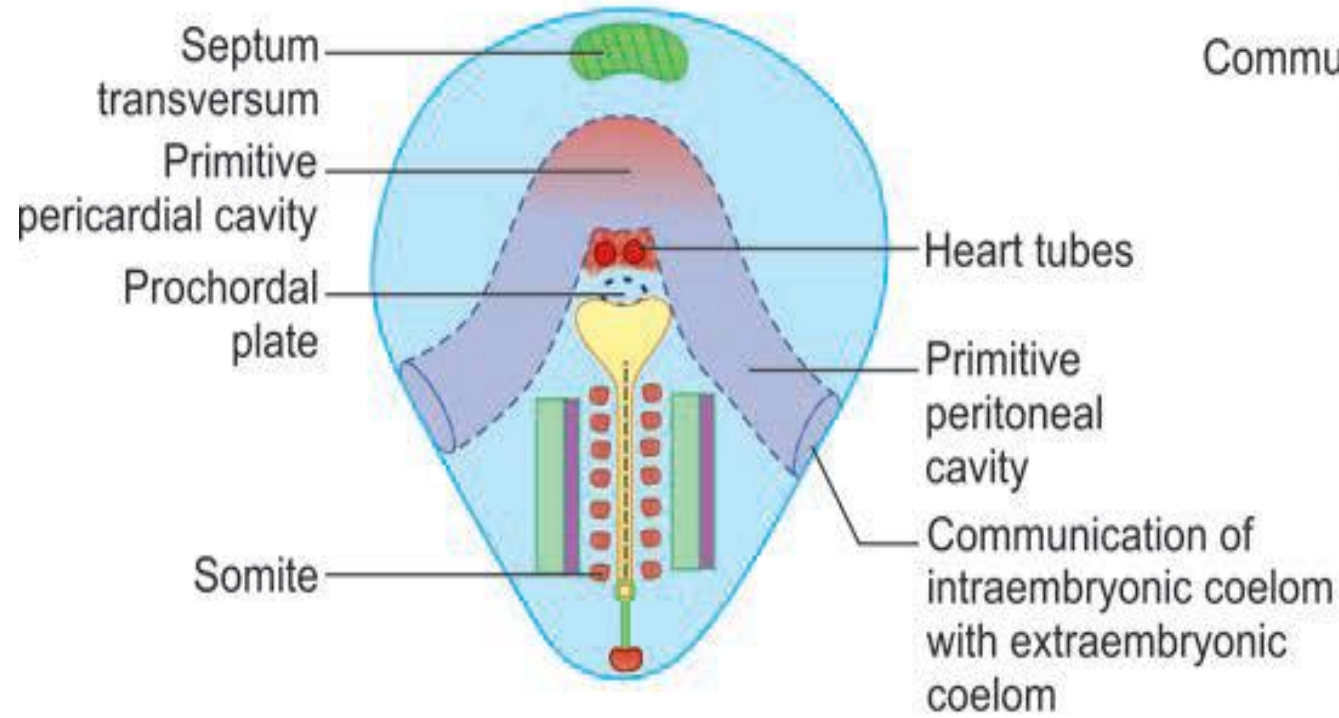
- **ALLANTOIC DIVERTICULUM**

- Before the formation of the tail fold, a small endodermal diverticulum called the *allantoic diverticulum* arises from the yolk sac near the caudal end of the embryonic disc
- diverticulum grows into the mesoderm of the connecting stalk
- With formation of the tail fold, part of this diverticulum is absorbed into the hindgut
- Passes from the ventral side of the hindgut into the connecting stalk

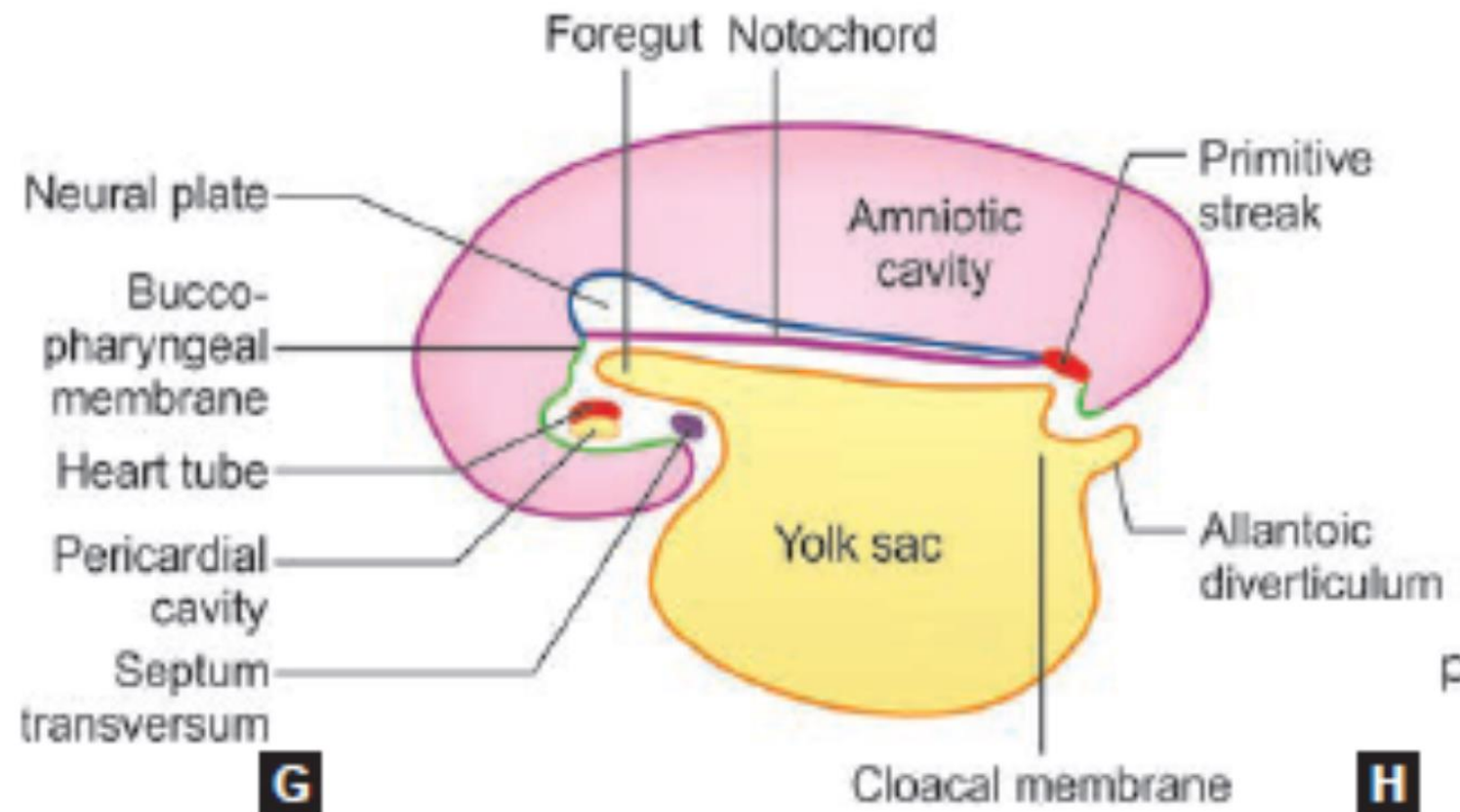


Figs 5.11A to F: Stages in the establishment of the umbilical cord, allantoic diverticulum and its relationship to the connecting

- **EFFECT OF HEAD AND TAIL FOLDS ON POSITIONS OF OTHER STRUCTURES**
- From the cranial to the caudal side, the structures seen in the midline are the:
- **Septum transversum**
- **Developing pericardial cavity and the heart**
- **Prochordal plate**
- **Neural plate**
- **Primitive streak**
- **Cloacal membrane**
- After folding, the relative positions of these structure change

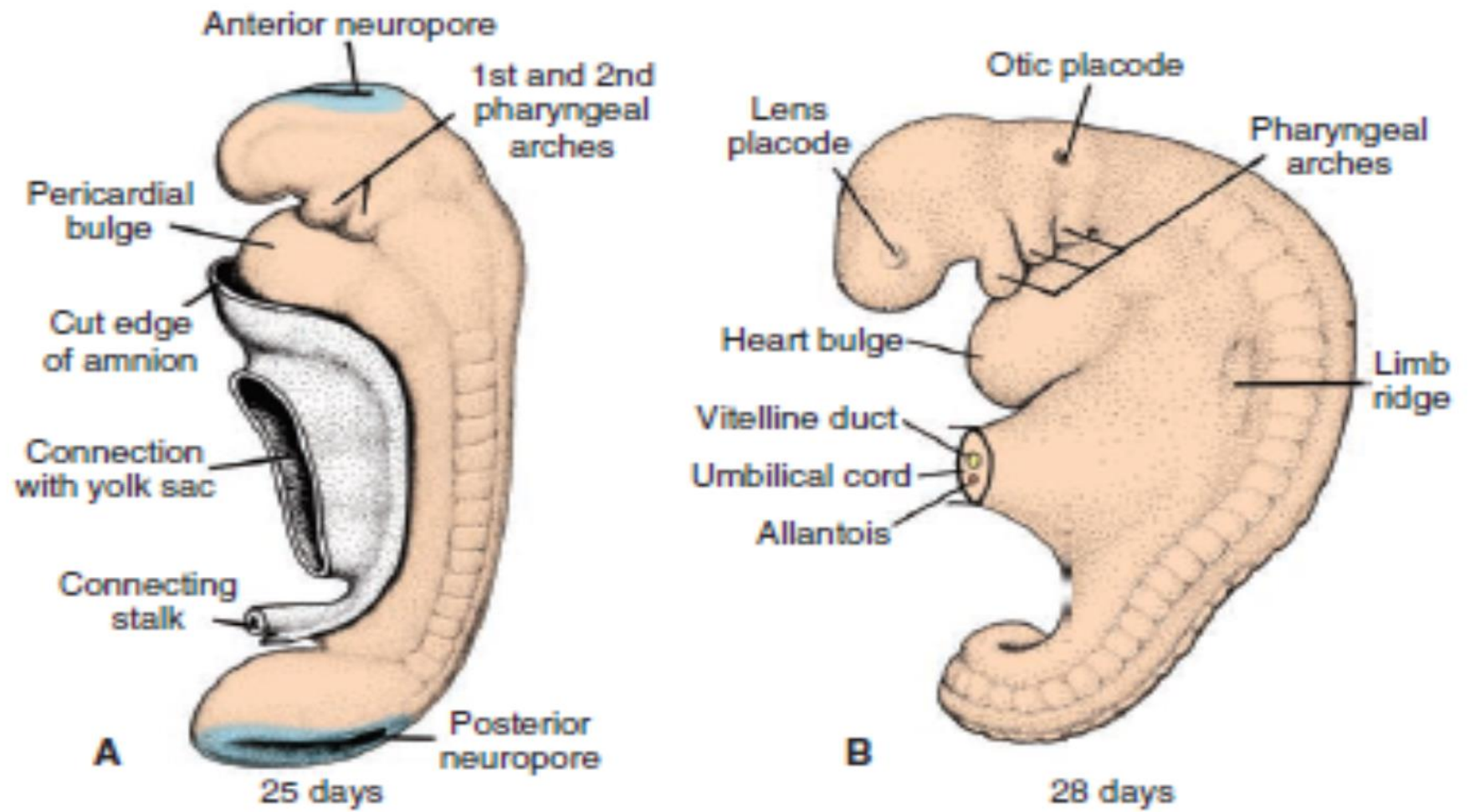


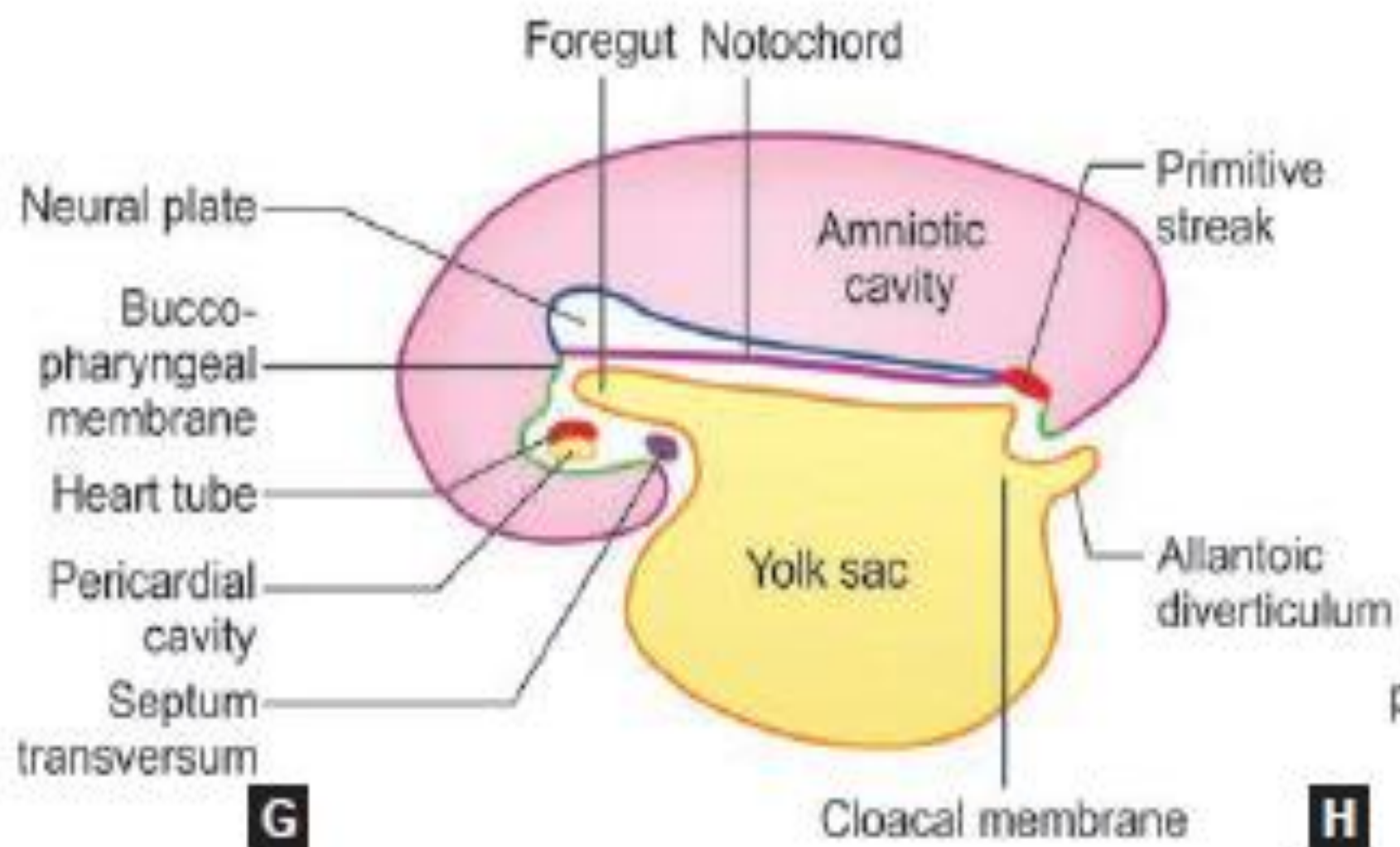
- With the formation of the head fold, it brings *heart & mouth* comes to lie on the ventral side of the embryo the eventual adult position.
- The pericardium enlarges rapidly and forms a conspicuous bulging on the ventral side of the embryo
- ***septum transversum***, which was the most cranial structure in the embryonic disc now lies caudal to the heart where it subsequently develops into the central tendon of the diaphragm



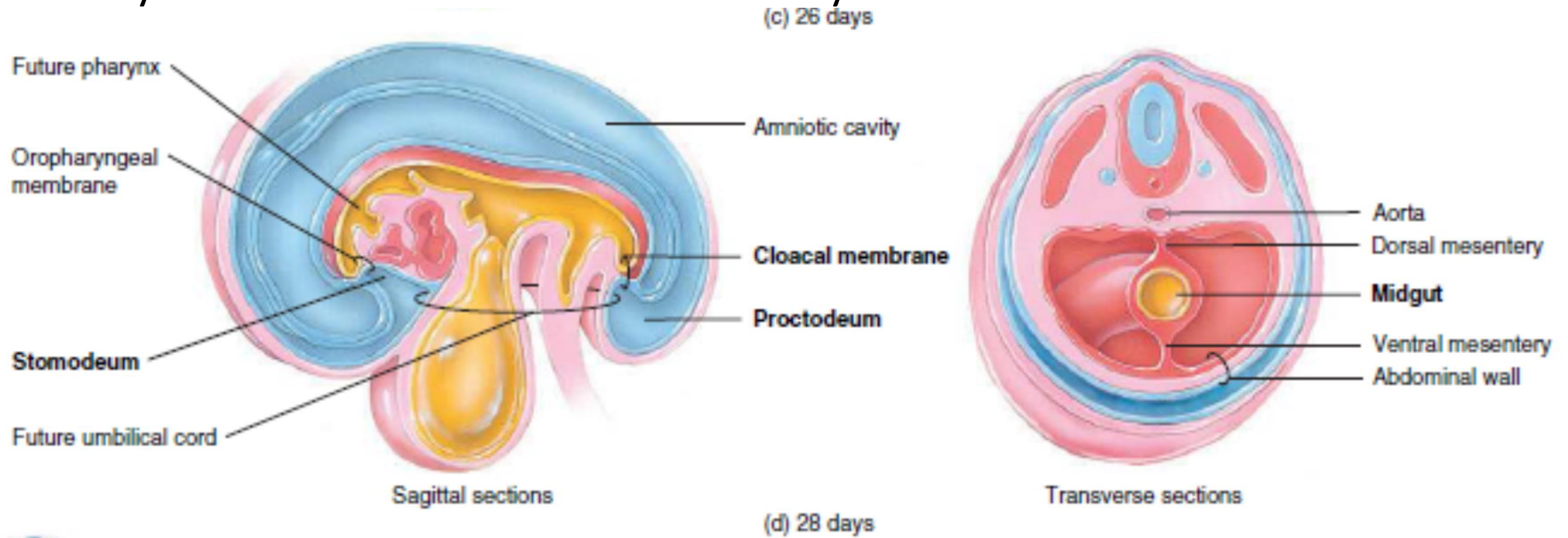
- most cranial structure of the embryo is now the enlarged cranial part of the neural tube, which later forms the brain
- There are now **2 big bulgings on the ventral aspect** of the embryo
- Cranially, there is the developing brain, and a little below it there is the bulging pericardium

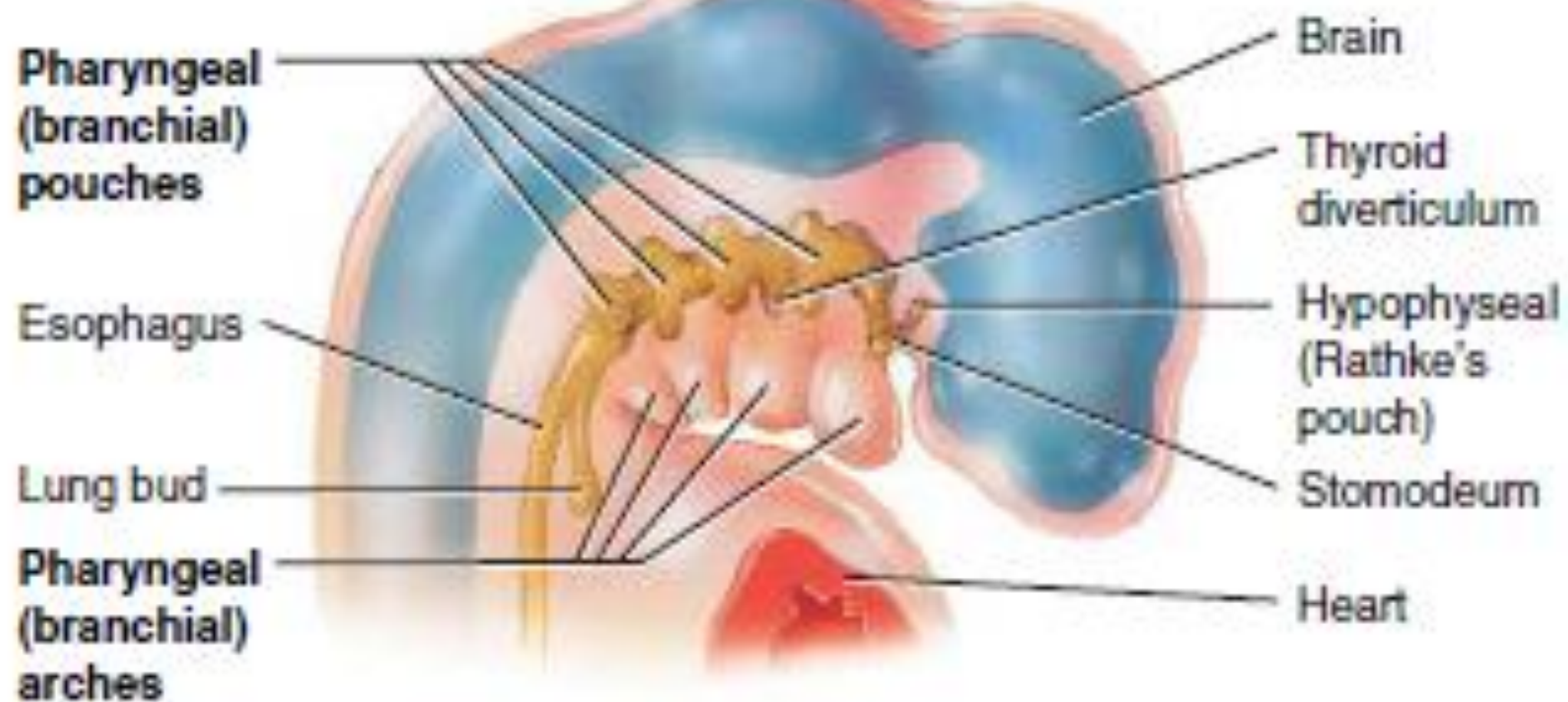
Chapter 6 Third to Eighth Weeks: The Embryonic Period



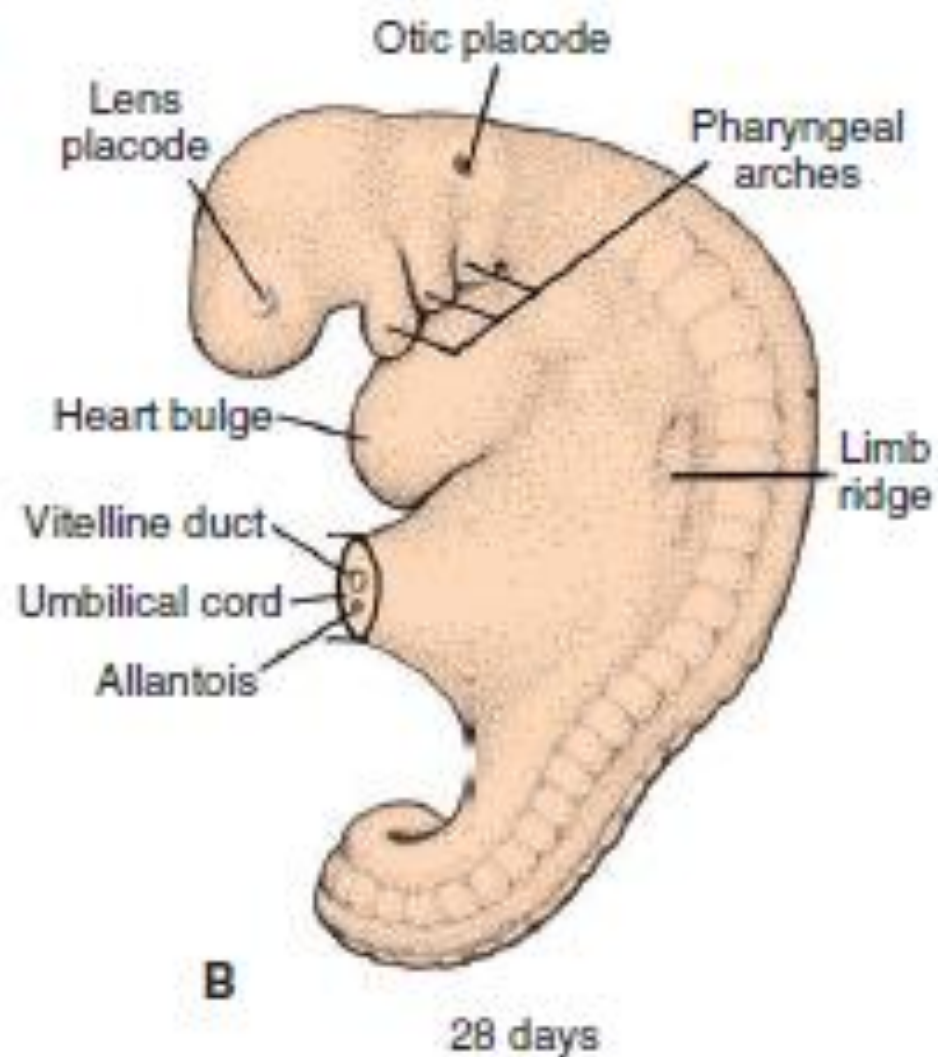
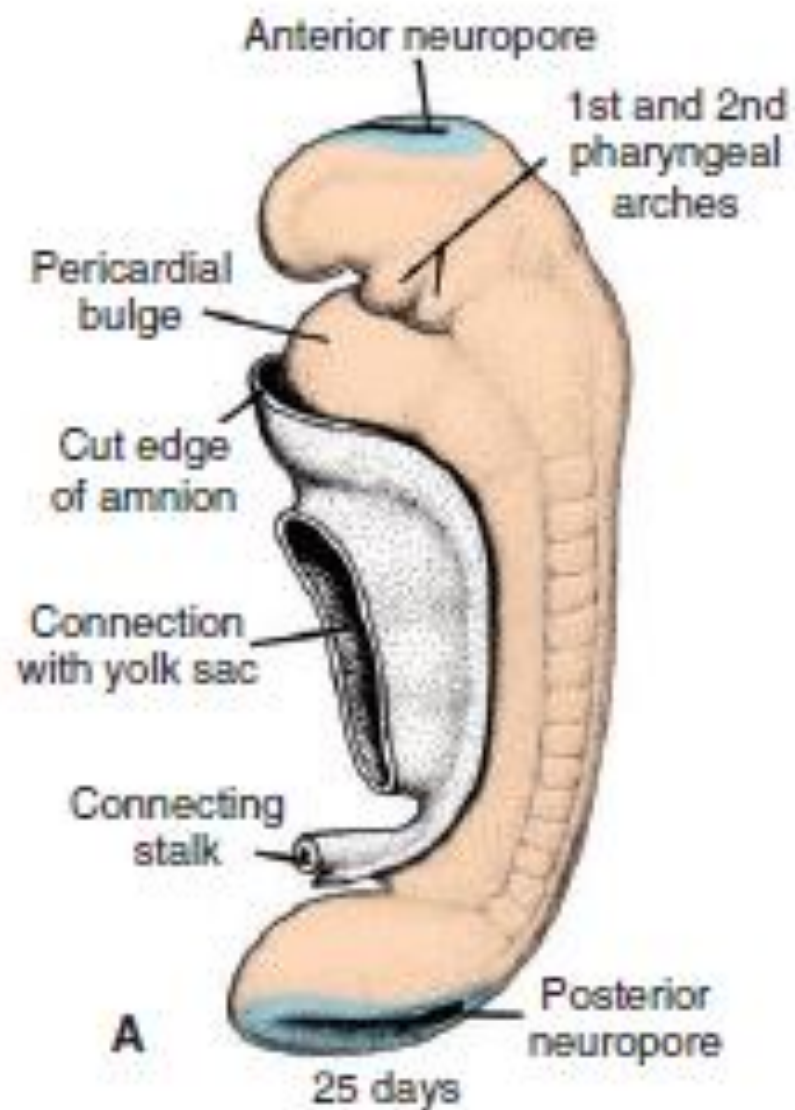


- In between developing brain & bulging pericardium, there is a depression called the *stomatodeum* or *stomodeum*, the floor of which is formed by the buccopharyngeal membrane
- Toward the tail end of the embryo, the primitive streak is now an inconspicuous structure that gradually disappears
- distal end of the hindgut is closed by the *cloacal membrane* first, this is directed caudally but later it comes to face ventrally





(b) Sagittal section, about 28-day embryo



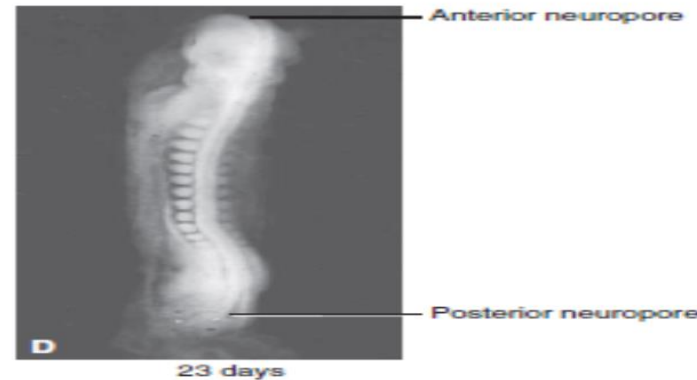
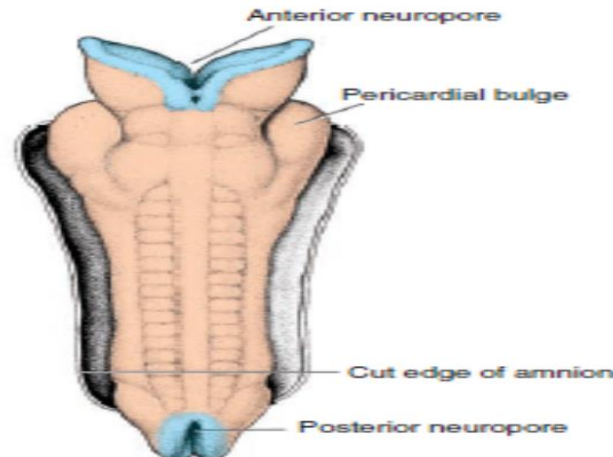
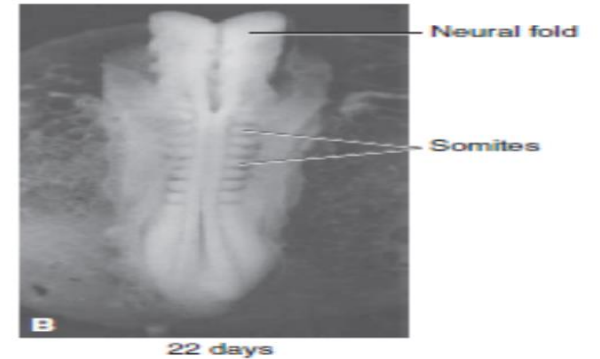
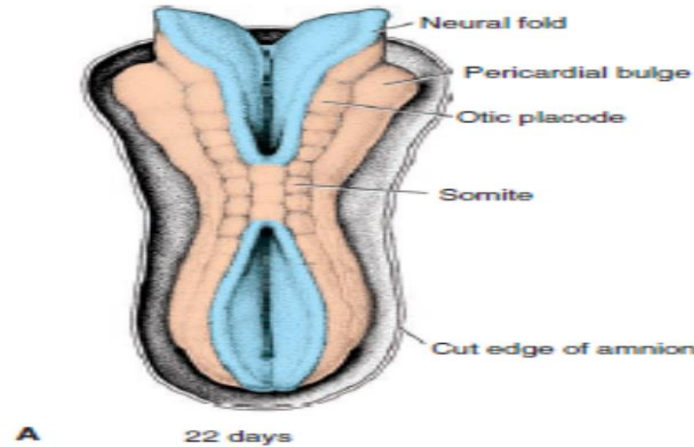
- **GERM LAYER DERIVATIVES**

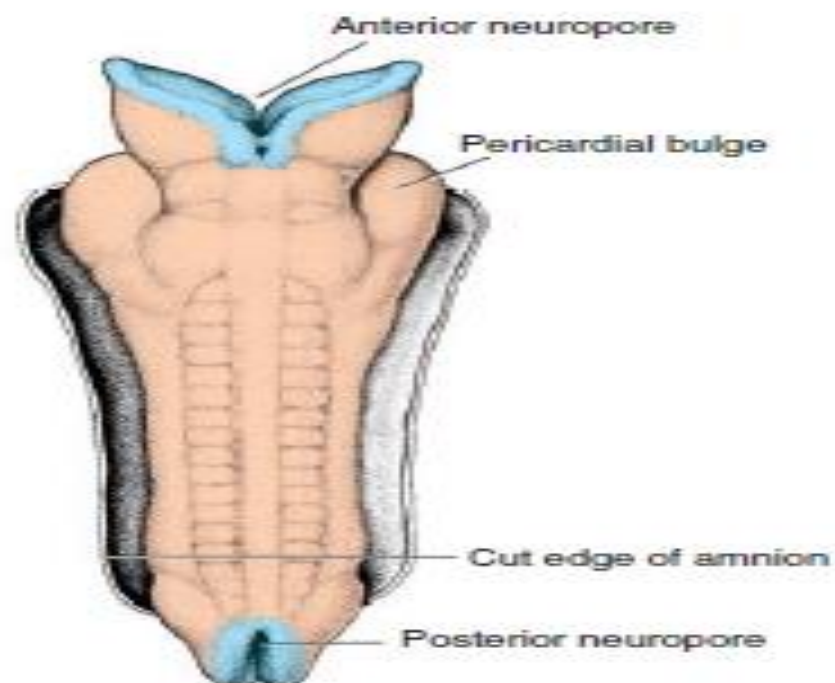
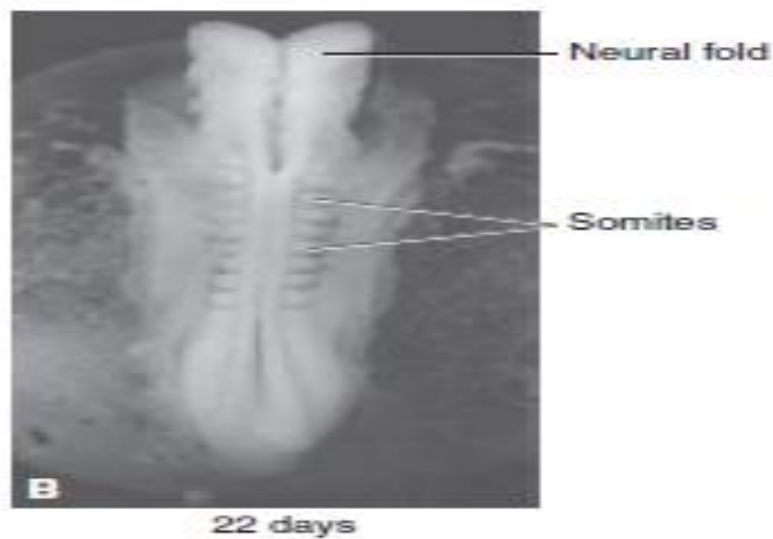
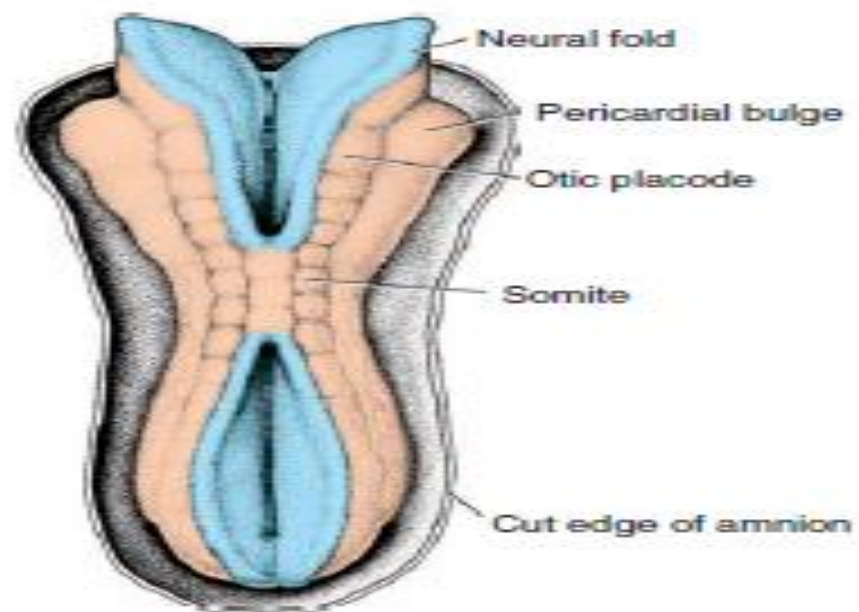
- 3 germ layers (ectoderm, mesoderm, and endoderm) formed during gastrulation give rise to the primordia of all the tissues and organs
- cells of each germ layer divide, migrate, aggregate, and differentiate in rather precise patterns as they form the various organ systems
- The main germ layer derivatives are as follows:
- **Ectoderm** gives rise to the:-
 - central nervous system, peripheral nervous system; sensory epithelia of the eye, ear, and nose; epidermis and its appendages (hair and nails);
 - mammary glands; pituitary gland; subcutaneous glands; and enamel of teeth
 -

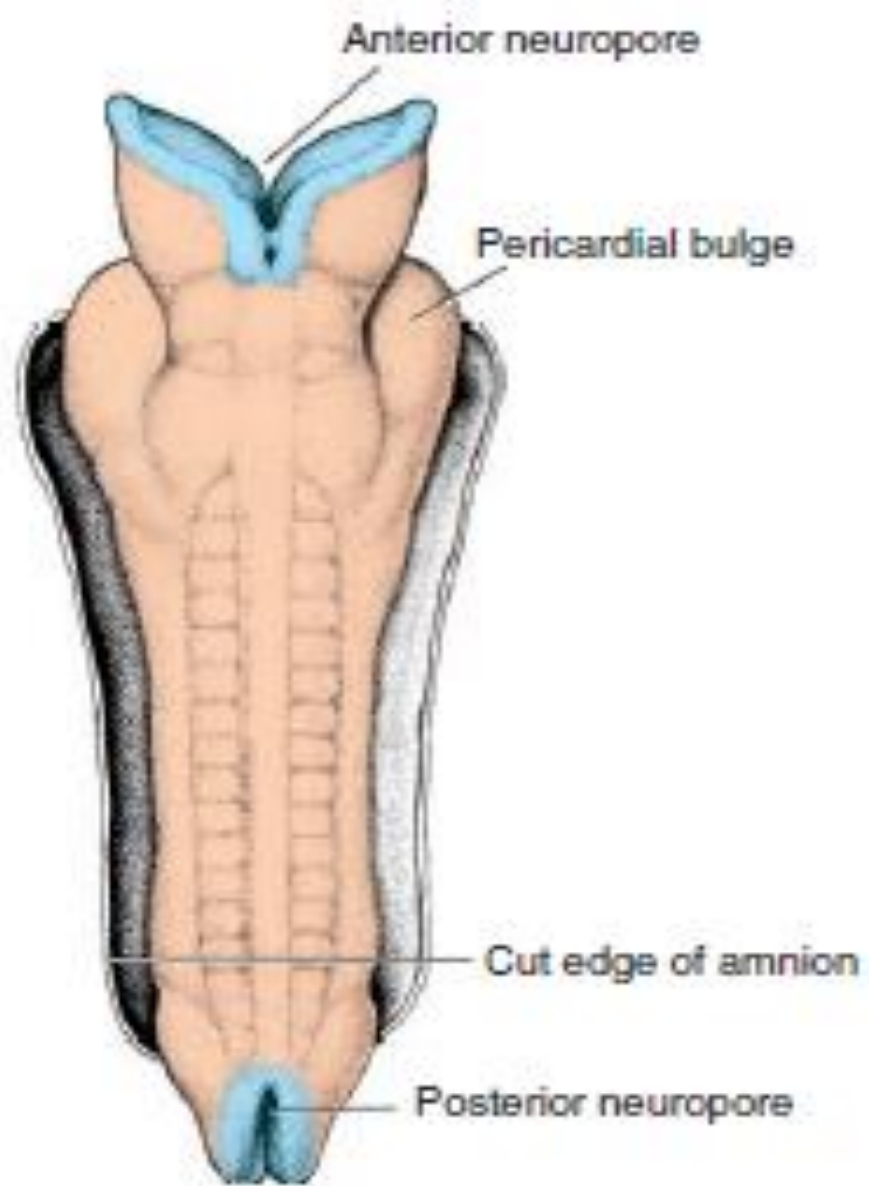
- **Neural crest cells**, derived from neuroectoderm, give rise to the :-
- cells of the spinal, cranial (cranial nerves V, VII, IX, and X), and autonomic ganglia; ensheathing cells(schwann cells) of the peripheral nervous system
- pigment cells(dermis); muscle, connective tissues, and bone of pharyngeal arch origin; suprarenal medulla; and meninges (coverings) of the brain and spinal cord
- **Mesoderm** gives rise to
- connective tissue; cartilage; bone; striated and smooth muscles; heart, blood, and lymphatic vessels; kidneys; ovaries
- testes; genital ducts; serous membranes lining the body cavities (pericardial, pleural, and peritoneal); spleen; and cortex of suprarenal glands

- **Endoderm** gives rise to the
- epithelial lining of the gastrointestinal and respiratory tracts, parenchyma of the tonsils, thyroid and parathyroid glands
- thymus, liver, and pancreas, epithelial lining of the urinary bladder and most of the urethra, and the epithelial lining of the tympanic cavity, tympanic antrum, and pharyngotympanic (auditory) tube

- **HIGHLIGHTS OF THE FOURTH TO EIGHTH WEEKS**
- Major changes in body form occur during the 4th week
- At the beginning, embryo is almost straight & has 4 to 12 somites that produce conspicuous surface elevations
- **neural tube** is formed opposite the somites, but it is widely open at the rostral and caudal **neuropores**

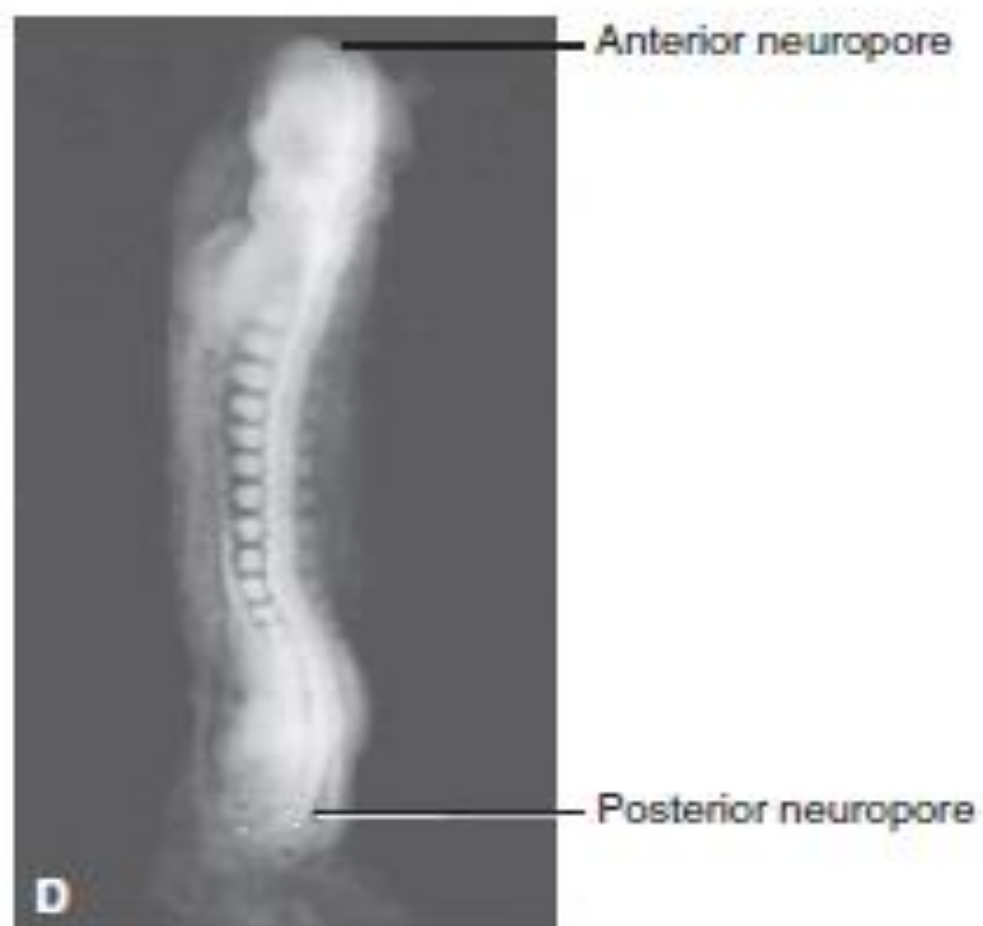






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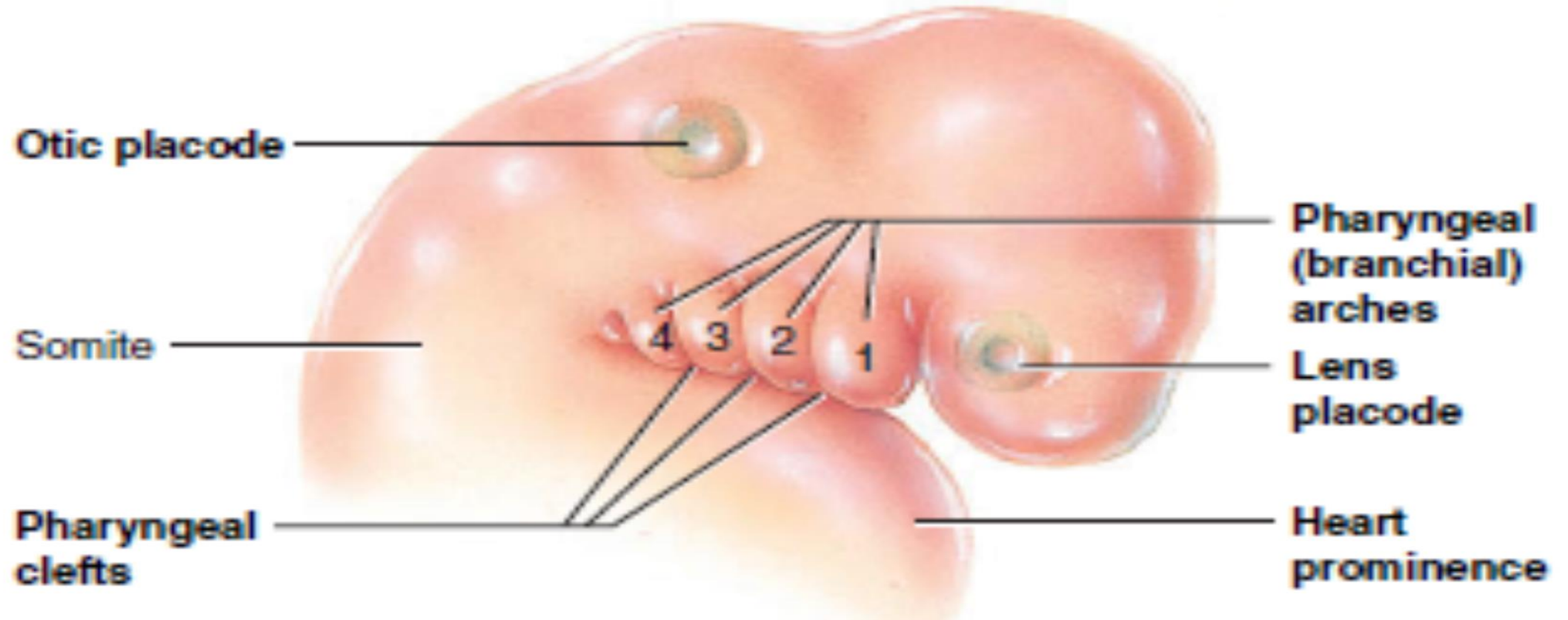
23 days



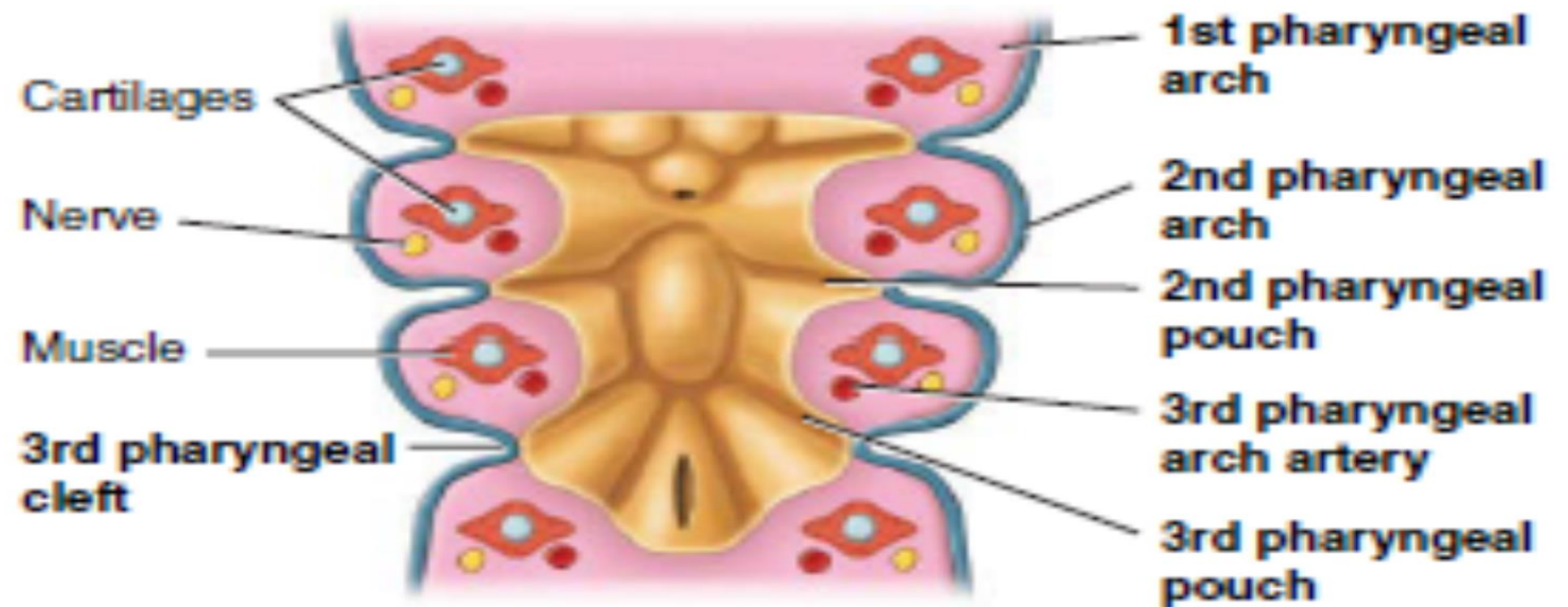
D

23 days

- 4 pairs of **pharyngeal arches** or *branchial arches* begin to develop on each side of the future head & neck regions during the 4th week
- form swellings on the surface of the embryo
- By 24 days, the first 2 pharyngeal arches are visible
- 1st (mandibular arch) & 2nd (hyoid arch) are distinct
- 3 pairs of **pharyngeal arches** are visible by 26 days

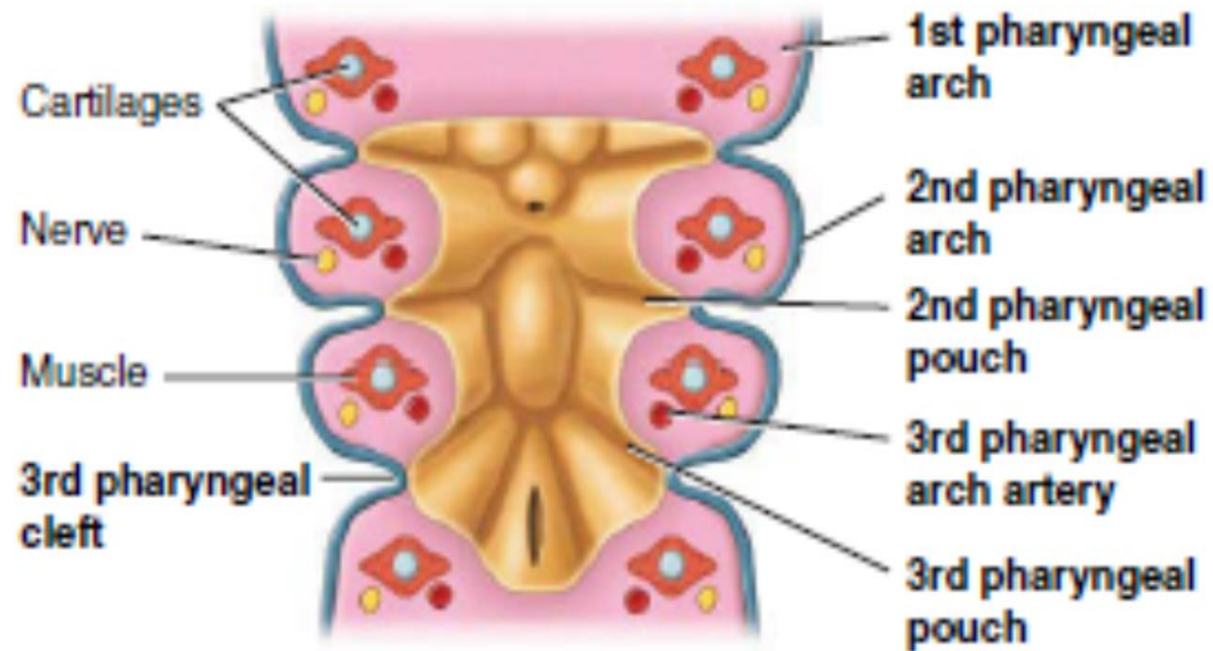


- Each pharyngeal arch consists of an outer covering of ectoderm & an inner covering of endoderm, with mesoderm in between
- Within each pharyngeal arch there is:-
- **an artery, a cranial nerve, cartilaginous skeletal rods** that support the arch & **skeletal muscle tissue** that attaches to and moves the cartilage rods
- On the ectodermal surface of the pharyngeal arch, each pharyngeal arch is separated by a groove called a **pharyngeal cleft** and endodermal surface by a **pharyngeal pouch**



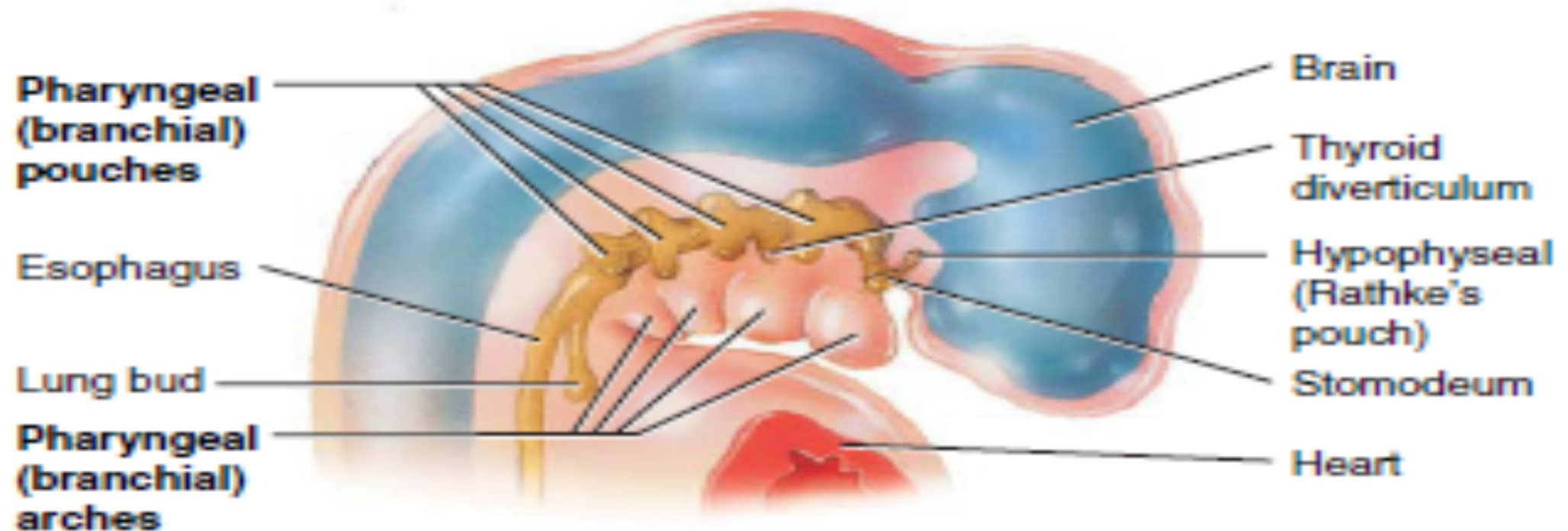
(c) Transverse section of the pharynx, about 28-day embryo

- Where the pharyngeal cleft & pouch meet to separate the arches, the outer ectoderm of the cleft contacts the inner endoderm of the pouch and there is no mesoderm between
- each pharyngeal arch, cleft, and pouch gives rise to specified structures in the head and neck
- Each pharyngeal arch is a developmental unit and includes a skeletal component, muscles, a distinct cranial nerve, and blood vessels.



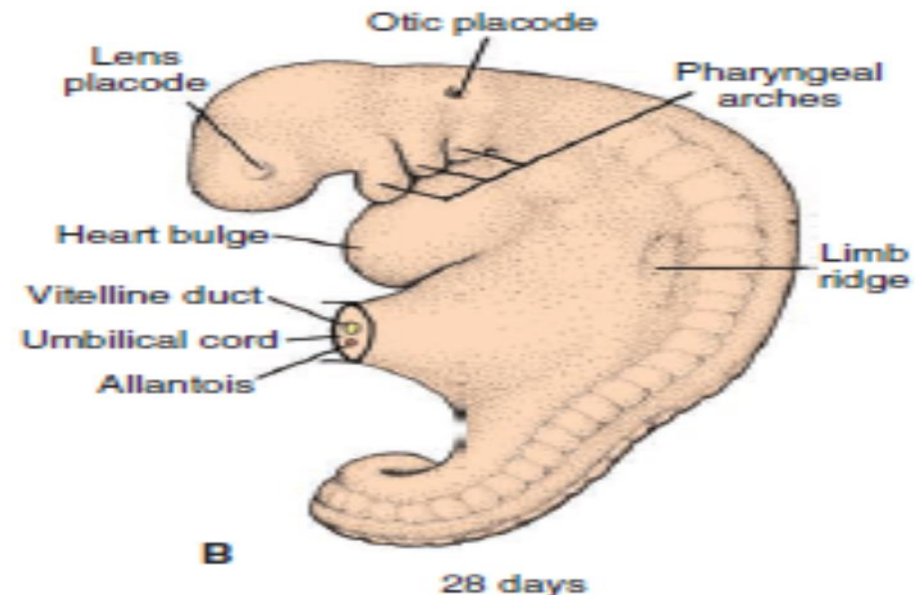
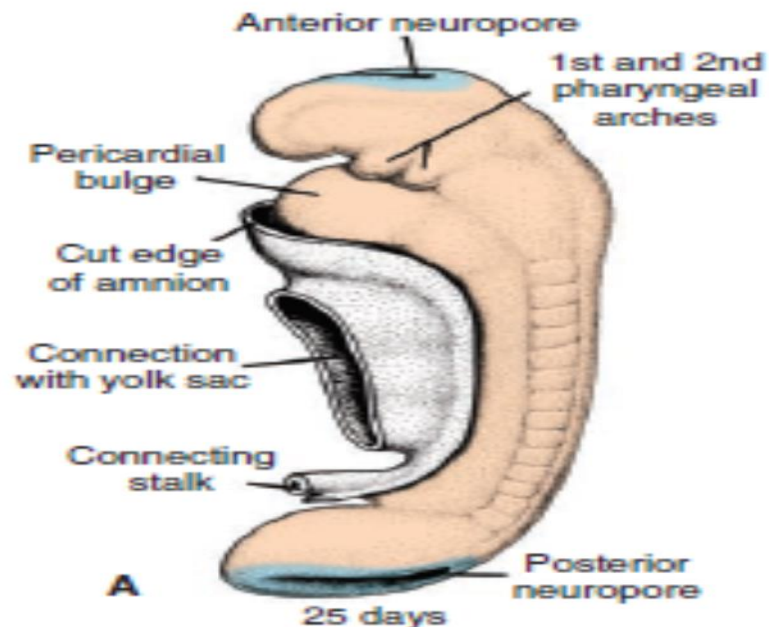
(c) Transverse section of the pharynx, about 28-day embryo

- the human embryo, there are four obvious pharyngeal arches
- Each of these arches develops into a specific and unique component of the head and neck region.
- For example, major part of the first arch gives rise to the:
- mandible (lower jaw), and a rostral extension of the arch, the maxillary prominence, contributes to the maxilla (upper jaw)

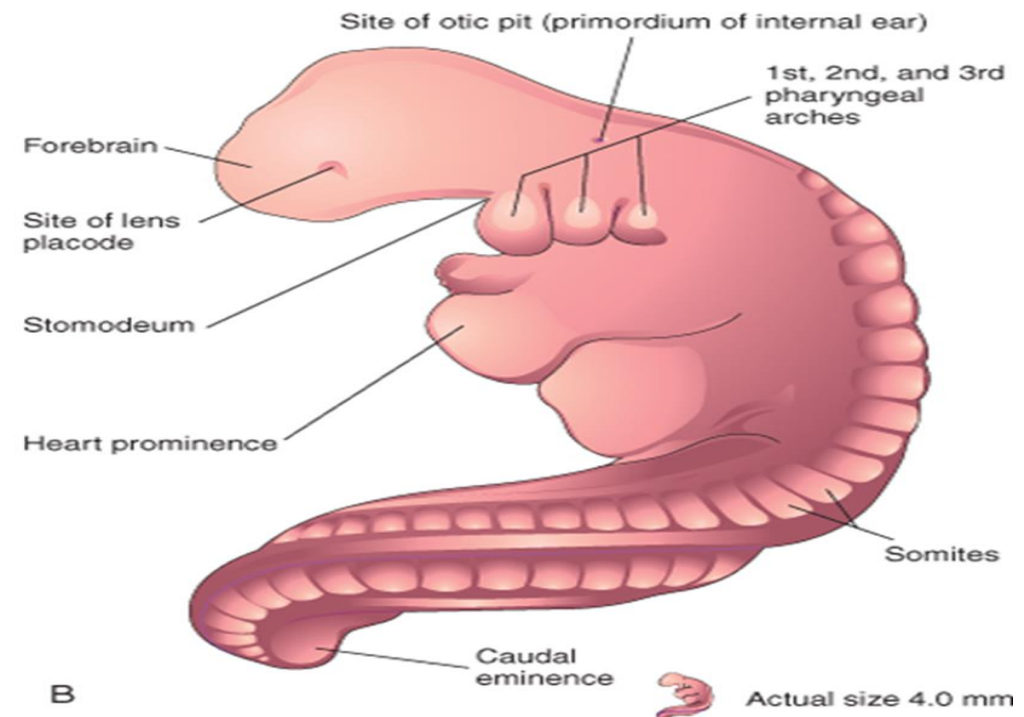


(b) Sagittal section, about 28-day embryo

- Day 26 the rostral neuropore is closed
- **forebrain** produces a prominent elevation of the head & folding of the embryo has given the embryo a C-shaped curvature
- **Upper limb buds** are recognizable by day 26 or 27 as small swellings on the ventrolateral body walls
- upper limbs begin their development as ectoderm-covered outgrowths
- first sign of a developing ear is a thickened area of ectoderm, the **otic placode** (future internal ear), which can be distinguished about 22 days after fertilization

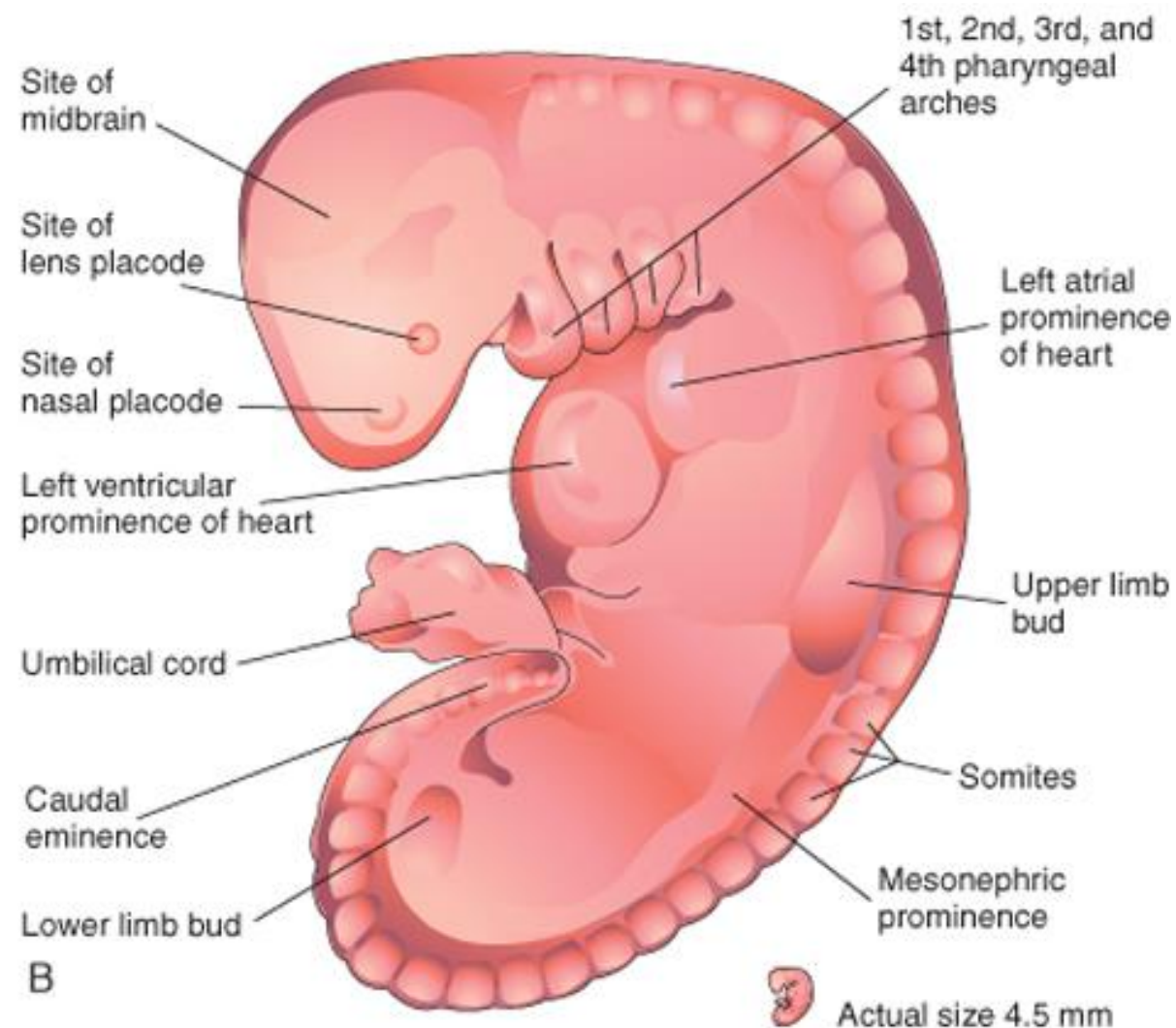


- A thickened area of ectoderm the **lens placode** which will become the eye, appears at this time
- By the end of the 4th week, the **lower limb buds** develop
- heart also forms a distinct projection on the ventral surface of the embryo called the **heart prominence, just** caudal to the pharyngeal arches
- a **tail** becomes visible the **caudal eminence**
- *By the end of the fourth week, the caudal neuropore is usually closed*



- **Fifth Week**

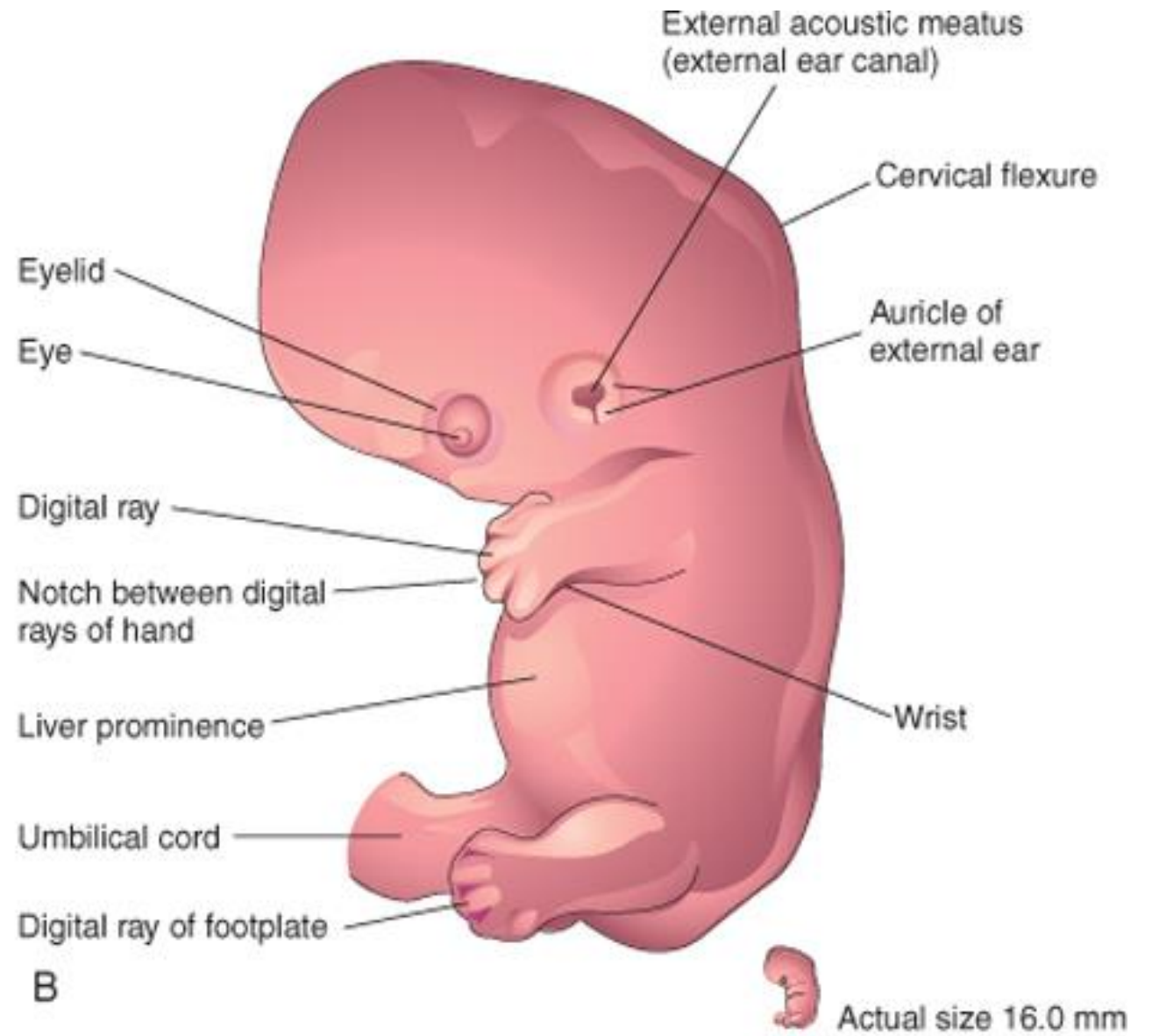
- Changes in body form are minor , but growth of the head exceeds that of other regions
- Enlargement of the head is caused mainly by the rapid development of the brain and facial prominences
- face soon contacts the heart prominence
- rapidly growing second pharyngeal arch overgrows the third and fourth arches, forming a lateral ectodermal depression on each side-the **cervical sinus**
- **Mesonephric ridges** indicate the site of the mesonephric kidneys, which are interim excretory organs



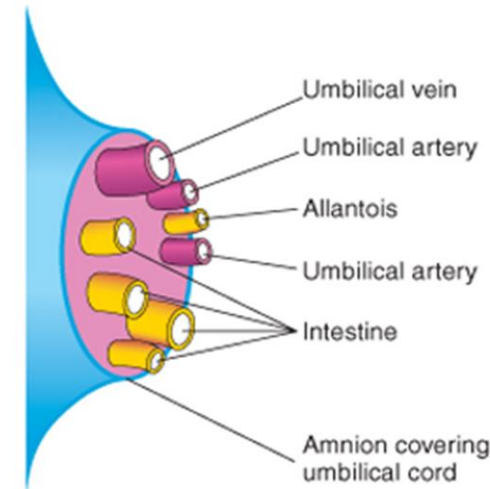
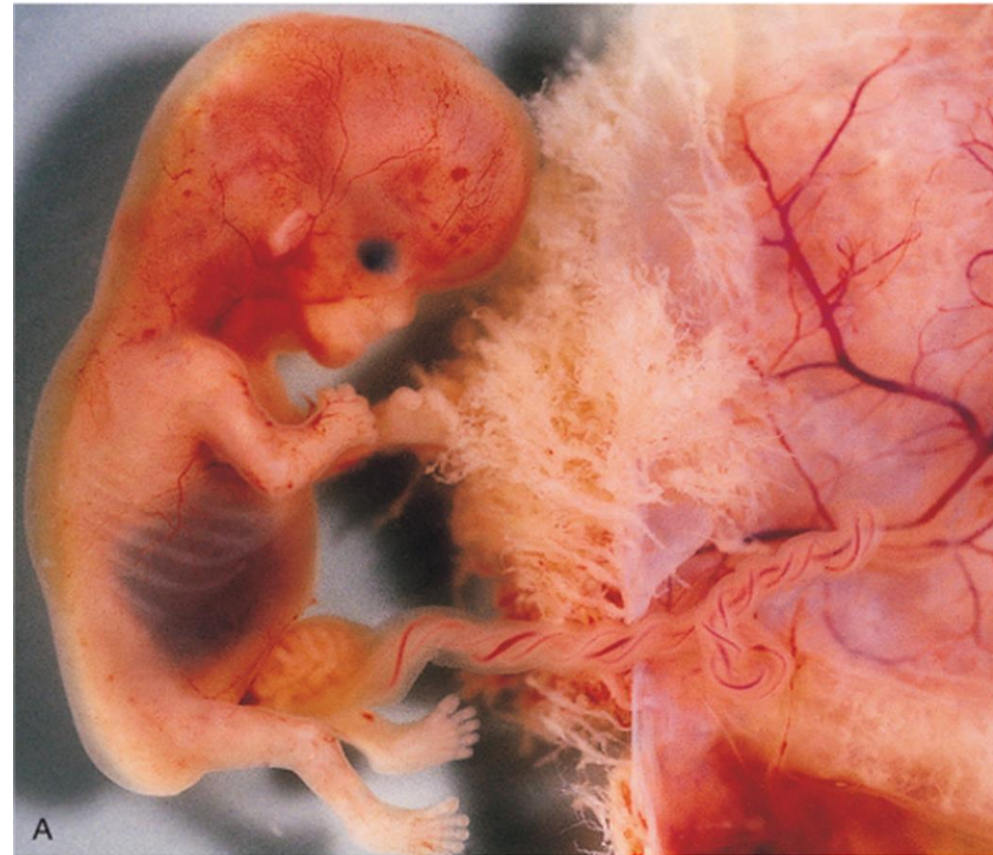
- **Sixth Week**

- embryos show reflex response to touch
- upper limbs begin to show **regional differentiation** as the elbows and large **handplates** develop
- primordia of the digits (fingers), called **digital rays**, begin to develop in the handplates, which indicate the formation of digits
- Embryos in the sixth week show **spontaneous movements**, such as twitching of the trunk and limbs
- Development of the lower limbs occurs 4 to 5 days later than that of the upper limbs

- Several small swellings-**auricular hillocks**-develop around the pharyngeal groove or cleft between the first two pharyngeal arches
- groove becomes the **external acoustic meatus** (external auditory canal)
- auricular hillocks contribute to the formation of the auricle, the shell-shaped part of the external ear
- retinal pigment has formed, the eye is now obvious
- head is now much larger relative to the trunk and is bent over the **heart prominence**.
- head position results from bending in the cervical (neck) region



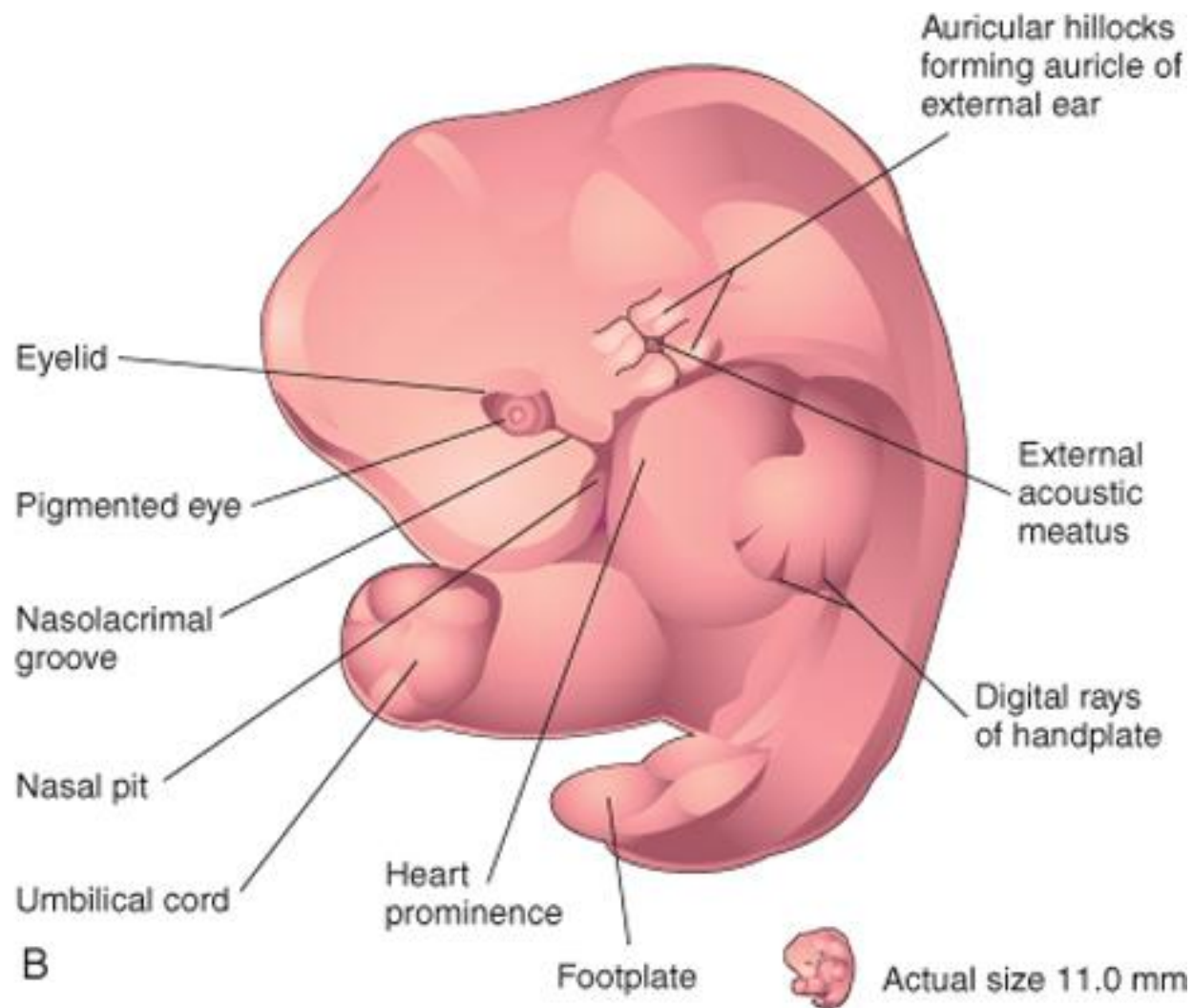
- **intestines** enter the extraembryonic coelom in the proximal part of the umbilical cord
- umbilical herniation is a normal event in the embryo
- herniation occurs because the abdominal cavity is too small at this age to accommodate the rapidly growing intestine



- **Seventh Week**
- limbs undergo considerable change during the seventh week
- **Notches** appear between the **digital rays in the handplates**, clearly indicating the future digits
- Communication between the primordial gut and umbilical vesicle is now reduced to a relatively slender duct, the omphaloenteric duct
- By the end of the seventh week, **ossification of the bones of the upper limbs** has begun

- **Eighth Week**

- final week of the embryonic period, the digits of the hand are separated but noticeably webbed
- Notches are now clearly visible between the digital rays of the feet
- **scalp vascular plexus** has appeared and forms a characteristic band around the head
- By the end of the eighth week, all regions of the limbs are apparent, the digits have lengthened
- are completely separated no longer webbed due to removal of cells via apoptosis
- the embryo has distinct human characteristics



- the head is still disproportionately large, constituting almost half of the embryo
- neck region is established, and the eyelids are more obvious
- eyelids are closing & by the end of 8th week, they begin to unite by epithelial fusion
- intestines are still in the proximal portion of the umbilical cord
- auricles of the external ears begin to assume their final shape
- external genitalia not distinctive enough to permit accurate sexual identification

- **Clinical Application**

- Knowledge of embryonic age is important because it affects clinical management, especially when invasive procedures such as chorionic villus sampling and amniocentesis are necessary
- In some women, estimation of gestational age from the menstrual history alone may be unreliable
- probability of error in establishing the LNMP (Last Normal Monthly Period) is highest in women who become pregnant after cessation of oral contraception because the interval between discontinuance of hormones and the onset of ovulation is highly variable
- In others, slight uterine bleeding ("spotting"), which sometimes occurs during implantation of the blastocyst, may be incorrectly regarded by a woman as light menstruation

- Other contributing factors to LNMP unreliability may include oligomenorrhea (scanty menstruation), pregnancy in the postpartum period (i.e., several weeks after childbirth), and use of intrauterine devices
- Despite possible sources of error, the LNMP is a reliable criterion in most cases